

ORIGINAL ARTICLE

Stock price forecasting through symbolic dynamics and state transition graphs with a convolutional recurrent neural network architecture

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Received: 28 December 2024 / Accepted: 2 May 2025 / Published online: 31 May 2025

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Abstract

Accurate stock price forecasting remains a critical challenge in financial analytics due to volatile market conditions, non-stationary dynamics, and abrupt regime shifts that often defy traditional modeling techniques. This study proposes a comprehensive framework for stock price forecasting that integrates symbolic dynamics, graph-based state representations, and deep learning. By converting continuous-valued stock prices into discrete symbolic states representing amplitude and trend information, the method constructs transition matrices capturing probabilistic relationships within financial time series. These transition matrices are then processed by a convolutional recurrent neural network (CRNN), in which convolutional layers isolate local spatial dependencies in the symbolic-state domain, while recurrent LSTM layers capture multi-scale temporal dynamics extending across multiple time horizons. Experimental evaluations are conducted over prediction horizons of 1 day, 10 days, and 100 days, spanning pre-COVID, COVID, and post-COVID market regimes. The results indicate that while longer prediction horizons naturally incur greater forecasting uncertainty due to compounding variability, the integration of symbolic-state preprocessing with deep temporal modeling demonstrates significant robustness in handling non-stationary financial environments. During the stable pre-COVID period, the proposed methodology achieves reductions in mean squared error (MSE) of up to 98% relative to the volatile COVID phase, highlighting its capability to effectively leverage well-defined market patterns in stable economic conditions. Furthermore, the model consistently delivers competitive forecasting performance across all prediction horizons and market regimes. Collectively, these findings emphasize the potential of symbolic-state-based deep learning architectures as a viable pathway to address the complexity and volatility characteristic of modern financial markets.

Keywords Symbolic dynamics · Stock price forecasting · Graph signal processing · Convolutional recurrent neural network · Non-stationary time series

1 Introduction

Forecasting stock prices is a central objective in financial analytics, underpinning decisions in risk management, policy formation, and strategic investment. The inherent complexity of financial time series, characterized by high volatility, non-stationary, and strong nonlinear dependencies, presents a continuous challenge to researchers and practitioners. Early methods for financial time series modeling provided structured statistical frameworks that offered a degree of interpretability and analytical rigor [1–6]. However, these classical models assume relatively stable market conditions and linear or near-linear relationships, limiting their capability to adapt to the complex,



evolving patterns that characterize modern financial environments. In response, researchers have increasingly turned to more flexible methodologies that can handle richer dynamics. Modern machine learning (ML) and deep learning (DL) approaches, including long short-term memory (LSTM) networks and convolutional neural networks (CNNs), have demonstrated considerable promise in modeling complex time series. Despite these advances, these methods often rely on extensive feature engineering, encounter difficulties in interpreting their internal representations, and may not fully exploit the underlying qualitative patterns that drive price movements.

A supplementary approach to comprehending non-stationary financial time series is offered by symbolic dynamics, a paradigm that has its origins in nonlinear dynamics and chaos theory. By transforming continuous-valued signals into sequences of discrete symbols, symbolic dynamics highlights structural characteristics and reduces noise, enabling researchers to focus on meaningful qualitative behaviors rather than raw numerical fluctuations. This symbolic representation aligns naturally with graph-theoretic approaches, allowing complex temporal interactions to be visualized and analyzed as state transition networks. Such integration between symbolic encodings and graph-based modeling has the potential to enhance interpretability, reveal hidden market regimes, and better guide the choice of predictive models.

To fully capitalize on the structured information provided by symbolic dynamics and its associated graph-based abstractions, it is imperative to integrate these representations seamlessly with state-of-the-art predictive modeling frameworks. By encoding raw financial time series into discrete symbolic states that preserve essential amplitude and directional characteristics, the process obviates the need for extensive domain-dependent feature engineering. Subsequently, representing state transitions as weighted graphs provides a mathematically rigorous substrate, akin to Markov transition matrices, from which complex temporal dependencies can be systematically extracted. Advanced machine learning and deep learning architectures, particularly those equipped with recurrent units, can then operate directly on these state-space representations to identify latent patterns, long-range correlations, and regime-switching behaviors that remain obscured in conventional continuous-valued signals. In doing so, symbolic encodings not only offer a dimensionality-reduced and noise-filtered input domain but also facilitate informed architectural decisions, hyperparameter tuning, and the selection of learning objectives, thereby enhancing the stability and adaptability of predictive models under non-stationary and rapidly evolving market conditions.

This study enhances the accuracy and robustness of stock price forecasting by combining the interpretive strength of symbolic dynamics with the modeling capabilities of graph-based state representations and advanced deep learning architectures. The proposed pipeline begins by symbolically encoding amplitude and directional information from raw financial time series, thereby distilling noisy price data into discrete states emphasizing meaningful temporal patterns. Unlike purely data-driven models that rely on raw numerical sequences, this symbolic transformation reduces overfitting by filtering out stochastic fluctuations while preserving essential structural patterns. State transition matrices are constructed from these symbolic encodings, capturing the probabilistic relationships that govern market dynamics. These matrices serve as a compact and interpretable representation of financial trends, facilitating the extraction of key transition probabilities that encode short-term fluctuations and long-range dependencies. These transition matrices are fed into a convolutional recurrent neural network (CRNN). This hybrid architecture harnesses convolutional layers to extract local spatial features and recurrent layers to capture long-term temporal dependencies. Employing convolutional filters, the model effectively identifies prominent structural motifs in transition matrices, while the recurrent layers ensure the retention of historical dependencies crucial for robust predictions. Departing from traditional deep learning models that treat financial time series as raw numerical inputs, this approach structures data into symbolic sequences and transition matrices, enabling the extraction of latent regime shifts and nonlinear dependencies that remain hidden in conventional forecasting methods. This methodological shift enhances predictive stability under market turbulence. It provides a degree of interpretability, as transitions between symbolic states can be analyzed to understand regime-switching behaviors and volatility clustering. By synergizing the principled abstraction of symbolic states with the adaptive learning capabilities of deep neural networks, the proposed framework delivers more resilient, context-aware forecasts that retain scalability. Additionally, this hybrid formulation facilitates

generalization across multiple assets and market conditions, reducing sensitivity to overfitting while maintaining adaptability to non-stationary financial environments. By integrating symbolic encodings and CRNN-based learning, the proposed framework effectively captures hierarchical temporal structures, allowing for microstructural pattern recognition and broader trend forecasting across varying time horizons.

The contributions of this work are threefold. First, an integrated symbolic-dynamics-to-CRNN pipeline is introduced, extending beyond conventional feature engineering to provide a more principled approach for highlighting essential temporal structures. This integration bridges the gap between purely statistical approaches and deep learning methods, offering a structured yet data-driven representation that retains interpretability while improving predictive accuracy. Second, the proposed method is empirically evaluated across multiple prediction horizons (1-day, 10-day, and 100-day) and distinct market regimes (pre-COVID, COVID, and post-COVID), demonstrating adaptability and resilience under shifting conditions. Through this extensive evaluation, the study systematically examines how different market structures influence transition probabilities and neural network performance, shedding light on the robustness of symbolic-state representations under varying volatility regimes. Unlike traditional econometric approaches that struggle with capturing nonlinear dependencies or deep learning models that lack interpretability, this approach provides both probabilistic insights and scalable learning capabilities, offering a balanced perspective for theoretical advancements and practical applications in financial forecasting.

The following sections begin with a review of relevant literature, tracing developments from traditional econometric approaches to advanced machine learning and deep learning architectures, followed by an introduction to symbolic dynamics methodologies. Next, the theoretical foundations of symbolic dynamics are outlined, along with a detailed explanation of data preprocessing and model construction steps. The empirical results are then analyzed, emphasizing the potential of symbolic-state-based modeling and CRNN architectures as a promising framework for developing more robust and scalable financial forecasting models.

2 Literature review

The field of stock price forecasting encompasses a wide spectrum of approaches, each designed to capture distinct facets of financial time series complexity. The first subsection examines foundational econometric models such as ARIMA and GARCH and their extensions via machine learning and fuzzy logic frameworks. These methods combine the statistical rigor of traditional econometrics with the flexibility of data-driven feature engineering and linguistic rule encoding. The second subsection focuses on deep learning techniques, including RNNs, LSTMs, and CNNs, which alleviate the dependence on predefined features by learning relevant representations directly from raw data. Recent enhancements, such as attention mechanisms and hybrid architectures, further improve robustness in non-stationary market environments. Finally, the third subsection introduces symbolic dynamics and graph-based methodologies, emphasizing their ability to capture underlying nonlinear structures, identify market regimes, and enhance transparency of collective trends. In synthesizing these three complementary threads, this literature review outlines the progression toward more adaptive, explainable, and high-performing stock price forecasting models.

2.1 Machine learning, statistical learning, and fuzzy logic-based stock price forecasting

Machine learning (ML) and statistical learning methodologies have long been a focal point of financial time series forecasting, with a substantial body of literature exploring their application to stock price prediction. Traditional approaches rooted in statistical econometrics, such as ARIMA and GARCH models, rely on well-defined statistical assumptions to characterize time-dependent structures and volatility patterns [4, 7–10]. These classical models are often employed as baseline predictors due to their mathematical rigor, interpretability, and well-understood theoretical properties [4–6, 8].

Building on the limitations of traditional econometric frameworks, researchers have increasingly embraced advanced ML techniques for more adaptive forecasting solutions. Concurrently, advanced ML algorithms, ranging from margin-based classifiers like SVMs [11, 12] to ensemble methods such as random forests [13] and gradient boosting machines [14, 15], have gained traction by leveraging high-dimensional feature spaces and flexible function approximators. These approaches integrate sophisticated feature engineering techniques (e.g., technical indicators, sentiment metrics, and regime-switching markers) to model linear and nonlinear dynamics in complex financial environments. They are frequently augmented by hyperparameter optimization strategies and cross-validation protocols to enhance generalization performance and mitigate overfitting, thus often outperforming classical models in short-term forecasting horizons [16–18]. As the literature attests, these machine learning methodologies effectively address short-horizon volatility, laying the groundwork for specialized hybrid systems that incorporate deep learning architectures and domain-specific heuristics.

In parallel with these theoretical and methodological developments, a growing number of empirical studies highlight the practical efficacy of advanced machine learning approaches in various financial contexts. Gu et al. [15] proposed a hybrid forecasting model combining empirical wavelet transform and gradient boosting decision trees to predict the settlement prices of London Metal Exchange Nickel, demonstrating its superiority over traditional EMD-based methods. Andrade and Cunha [16] proposed an XGBoost-based method for disaggregated retail demand forecasting, addressing inventory inaccuracies, promotional impacts, and structural changes. Using a large real-world dataset, their model outperformed the Base-Lift benchmark, improving forecasting accuracy by 26.72% and reducing stockouts and excess inventory, offering a scalable and automated solution for retail operations. These empirical advancements emphasize not only the enhanced performance of modern ML models but also the necessity of exploring alternative paradigms that can adapt to evolving market conditions.

As the complexity of financial markets continues to challenge conventional modeling paradigms, fuzzy logic-based frameworks emerge as a compelling alternative to purely quantitative approaches. Fuzzy logic-based approaches provide an alternative way that incorporates linguistic variables and membership functions, thereby accommodating uncertainty, ambiguity, and imprecision inherent in financial markets [19–21]. Such systems can codify human expert knowledge into interpretable rule sets, particularly important in volatile and rapidly shifting market conditions, while preserving robustness against noisy and partially observed data. Nevertheless, although these methods have demonstrated promise in capturing short-term patterns and rapidly evolving market microstructures, the literature consistently notes challenges in modeling long-term dependencies and adapting to structural changes in financial time series. Researchers continue to explore hybrid architectures, transfer learning approaches, and deep representation learning techniques aimed at overcoming these limitations and achieving more stable, reliable, and interpretable long-term forecasting performance [19–21]. In this vein, new lines of inquiry revolve around the synergy of fuzzy logic systems with other sophisticated machine learning and deep learning models, paving a route toward more adaptive and explainable predictive frameworks.

Empirical work further consolidates the role of fuzzy logic and hybrid modeling, presenting evidence-based improvements in forecasting accuracy and decision support capabilities. Jiang et al. [22] demonstrated the superior performance of machine learning algorithms, particularly deep learning, in predicting stock price crash risks by leveraging firm-specific characteristics, with a focus on profitability and value versus growth features, offering nuanced economic interpretations and highlighting significant applications within state-owned enterprises and low economic policy uncertainty periods in the Chinese stock market. Chen et al. [21] proposed a fuzzy time-series model incorporating the Fibonacci sequence to improve stock price forecasting accuracy. Using TSMC and TAIEX stock data, the model outperformed existing fuzzy models by leveraging a fluctuation-weighted method, spread-center defuzzification, and Fibonacci-based forecasting adjustments, achieving significantly lower prediction errors. Hadavandi et al. [20] proposed a hybrid model combining genetic fuzzy systems (GFS) and self-organizing map (SOM) neural networks for stock price forecasting. The approach uses stepwise regression for feature selection, SOM for clustering, and GFS for rule extraction and database tuning. Applied to stock price data from IT and airline sectors, the model achieved superior forecasting accuracy compared to existing methods, as measured by mean absolute percentage error. Chang et al. [19] proposed a hybrid model

combining a Takagi–Sugeno fuzzy rule-based system and support vector regression (SVR) for stock trading forecasting. The approach dynamically identifies buy-sell trading thresholds using fuzzy sub-models trained on historical technical indicators. By utilizing piecewise linear representation for segmenting stock prices and SVR for forecasting trading signals, the model demonstrated superior profitability and stability compared to traditional regression and neural network-based methods in experiments. Kao et al. [23] introduced a hybrid stock price forecasting model integrating nonlinear independent component analysis (NLICA) for feature extraction and SVR for prediction. Using datasets from the Shanghai Stock Exchange Composite and Nikkei 225 indices, the approach demonstrated improved accuracy compared to models combining principal component analysis (PCA) or linear ICA with SVR. By extracting independent components to uncover hidden data patterns, the NLICA-SVR method effectively enhanced prediction performance, highlighting its potential for financial time series forecasting. Taken together, these studies highlight the tangible benefits of combining fuzzy logic principles with advanced ML techniques, reinforcing the notion that integrated approaches can yield more robust and context-aware forecasting tools.

In light of these diverse modeling strategies, ranging from classical econometric approaches to sophisticated ML and fuzzy logic frameworks, it becomes clear that no single method uniformly outperforms others across all market regimes and forecasting horizons. Each technique and combination thereof has inherent trade-offs in terms of complexity, computational burden, and capacity to capture nonlinearity and structural changes in financial time series. Importantly, interpretability remains a critical factor in financial decision making, as stakeholders must understand the rationale behind model predictions to ensure trust, compliance with regulatory standards, and alignment with institutional knowledge [24–27]. While deep learning architectures and hybrid models frequently deliver superior predictive performance, they often do so at the expense of interpretability, motivating continued research into explainable AI methods and transparent fuzzy rule extraction processes.

2.2 Deep learning-based stock price forecasting

Deep learning has revolutionized stock price forecasting by enabling end-to-end learning from raw data, eliminating the need for extensive feature engineering. Architectures such as recurrent neural networks (RNNs), long short-term memory (LSTM) networks, and convolutional neural networks (CNNs) excel at capturing temporal and spatial dependencies in financial time series. Recent advancements include hybrid models that integrate deep learning with attention mechanisms, reinforcement learning, and symbolic representations, enhancing their robustness in handling non-stationarity and complex market dynamics.

Empirical investigations further emphasize the practical benefits of deep learning. Jayanth Balaji et al. [28] conducted an empirical study on the applicability of various deep learning models, including LSTM, GRU, CNN, and ELM, for forecasting stock prices using the S&P BSE-BANKEX index data. Fourteen models were designed to generate one-day-ahead and four-day-ahead forecasts, evaluated on metrics like RMSE, median absolute percentage error, and directional accuracy. Results showed that GRU-based models excelled in short-term forecasts, while ELM models performed better for longer-term horizons. In a complementary line of research, Yu et al. [29] proposed a stock price forecasting model integrating locally linear embedding (LLE) for nonlinear dimensionality reduction with a backpropagation (BP) neural network. The study demonstrated that the LLE algorithm effectively reduces the dimensionality of high-dimensional, nonlinear financial data, improving prediction accuracy. Using stock data, the LLE-BP model was compared against BP, PCA-BP, and ARIMA models, showing superior performance in terms of RMSE and MAE. This method highlights its practical applicability for predicting stock prices in nonlinear and high-dimensional contexts.

Focusing on attribute selection and application-specific modeling, Laboissiere et al. [30] proposed a methodology using artificial neural networks (ANNs) to forecast the maximum and minimum daily stock prices of Brazilian power distribution companies traded on the Sao Paulo Stock Exchange. The approach combined attribute selection based on correlation analysis with ANNs to improve prediction accuracy. Key attributes, including historical prices, market indexes, and American dollar quotes, were used as inputs. The methodology

achieved low MAPE for maximum (below 0.9%) and minimum (below 2.1%) price predictions, demonstrating its effectiveness in guiding investment decisions in the electricity sector. Li et al. [31] introduced a novel framework, Chart GCN, which employs graph convolutional networks (GCNs) to analyze stock price movements by transforming technical charts into graphs, outperforming traditional technical analysis and baseline deep learning models with prediction accuracies of 69.26% on SZ-50 and 68.62% on CSI-300 datasets, while achieving the highest net trading values in simulations. Long et al. [32] developed a multi-filters neural network (MFNN) combining convolutional and recurrent units for financial time series feature extraction, achieving higher prediction accuracy compared to traditional machine learning and statistical models. Md et al. [33] proposed a novel approach for stock price forecasting using a multilayer sequential (MLS) LSTM model. The model uses normalized time-series data to capture long-term dependencies and overcome challenges like the vanishing gradient problem. Tested on Samsung stock data, the MLS-LSTM achieved 95.9% accuracy on training data and 98.1% accuracy on testing data, outperforming other machine learning and deep learning methods. With a mean MAPE of 2.18%, the method demonstrated highly accurate predictions, showcasing its potential as a practical tool for stock price forecasting. Tian et al. [34] presented a hybrid model combining a bidirectional LSTM network and Bayesian-optimized LightGBM for stock price prediction. Utilizing Pearson's correlation coefficient for feature selection, the model integrates LSTM for sequence learning and LightGBM for regression, with Bayesian optimization refining LightGBM parameters. Testing on ten stocks, including ES = F and YM = F, showed the hybrid model consistently outperforming alternatives such as single LSTM and RNN models, with improved RMSE, MAE, accuracy, and F1-score. The study demonstrates the model's superior approximation and generalization abilities for forecasting stock price fluctuations.

Compared to earlier statistical and machine learning methods that often depend on handcrafted features and explicit modeling assumptions, deep learning frameworks can learn complex patterns directly from raw data, thereby reducing the need for extensive expert-driven variable selection. The evolution from traditional econometric and ML methodologies to deep learning-based approaches marks a shift toward end-to-end learning, adaptability, and automatic feature extraction. While deep learning has yet to achieve the full interpretability and expert knowledge encoding capabilities seen in fuzzy logic systems and statistical learning approaches and often involves more complex and computationally expensive training processes, continued research into explainable deep learning, attention mechanisms, and hybrid frameworks promises to yield forecasting solutions that are both powerful and contextually meaningful.

2.3 Symbolic dynamics in time series analysis for stock price forecasting

Symbolic dynamics, originally developed in the fields of nonlinear dynamics and chaos theory, provides a framework for transforming continuous time series into discrete symbol sequences that preserve essential structural and dynamical characteristics [35, 36]. This emphasis on nonlinear phenomena makes it particularly suitable for analyzing the often nonlinear and chaotic behavior of stock price changes [1–3]. The symbolic dynamics approach allows complex, non-stationary signals, such as stock prices, to be analyzed through patterns of symbols rather than raw numerical values, thus offering a way to reduce noise and highlight underlying regimes of behavior. By mapping continuous data to a finite alphabet and analyzing the resulting symbolic sequences, researchers can detect shifts in market volatility, identify recurrent patterns, and characterize long-term dependencies that may not be easily discernible using classical methods or direct numerical processing [35, 37].

Building upon the foundation laid by symbolic dynamics, graph-based approaches in nonlinear signal processing further enhance our ability to characterize and interpret complex time series data [38]. By representing symbolic sequences as networks, where symbols correspond to nodes and transitions between them form edges, these methods translate temporal patterns into topological structures that can be examined using tools from graph theory. Notably, these graph-based techniques have found applications across a wide range of natural and engineered systems, including vibration analysis in mechanical structures [37], biomedical signal processing [39], and other domains where uncovering hidden patterns is crucial. Networks generated from symbolized financial

time series, for example, can capture transitions between volatility regimes that might indicate impending market shifts, reveal community structures corresponding to various market states, and offer a glimpse into the resilience and robustness of price dynamics.

Wang et al. [40] review graph-based methods for stock price prediction, focusing on graph construction and learning. They discuss the importance of incorporating relational information and external factors, such as economic indicators and social media, into stock market models. The study emphasizes constructing explicit and implicit stock-to-stock relationships and explores specialized graph types like heterogeneous graphs, hyper-graphs, and subgraphs. Additionally, feature extraction methods, including graph embeddings and neural networks, are highlighted. The review underscores the potential of graph learning to improve stock market predictions by effectively modeling the complex relationships and dynamics within financial markets. Jafari and Haratizadeh [41] introduced GCNET, a graph-based framework for predicting stock price movements. GCNET models the relationships among stocks using an influence network and employs graph convolutional networks (GCNs) in a semi-supervised learning approach. It assigns initial labels to a subset of stocks using history-based predictors and refines predictions by leveraging information from graph structures. Experiments on NASDAQ stocks demonstrated GCNET's superior performance compared to state-of-the-art methods, highlighting its ability to model complex inter-stock relationships and improve prediction accuracy. Song et al. [42] introduced the MGAR model to improve stock return ranking predictions by leveraging dynamic and multi-relational graphs. It incorporates a price similarity relation graph alongside industry and Wiki relation graphs to capture evolving stock interactions. Using a multi-relational graph attention mechanism and LSTM-based feature extraction, MGAR effectively combines temporal and relational embeddings for accurate predictions. Experiments on NASDAQ and NYSE datasets show MGAR achieves superior returns and ranking accuracy compared to advanced methods, demonstrating its robustness in modeling complex stock relationships. Long et al. [43] proposed the Deep Stock-trend Prediction Neural Network (DSPNN), integrating trading records and public market data to predict stock price trends in the Chinese stock market. By leveraging knowledge graphs to identify correlations between stocks, clustering investors to analyze trading patterns, and combining convolutional neural networks with attention-based BiLSTMs, the model achieves enhanced accuracy for predicting short-term stock movements and trends. Experimental results demonstrate the robustness and effectiveness of the approach, with accuracies exceeding 70% for predicting trends over multiple trading days. Wang et al. [44] proposed a novel model integrating knowledge graph (KG), GCN, and community detection for large-scale stock price prediction. The authors construct a stock KG to capture explicit and implicit relationships between stocks and incorporate a GCN-based community detection approach to refine clusters of similar stocks. These clusters serve as input features for time-series models (LSTM and GRU), enabling improved trend prediction. Evaluated on a dataset of 762 stocks from China's A-share market (2013–2019), the model achieves superior accuracy, precision, and stability compared to state-of-the-art methods, addressing challenges like dataset bias and prediction instability in large-scale settings. Wu et al. [45] introduced a novel stock prediction framework leveraging complex network theory by transforming time series data into price graphs to address challenges such as long-term dependencies and chaotic properties in financial time series. Using structural information extracted from graph embeddings and node weights, their framework effectively captures both short- and long-term dependencies. Enhanced by attention mechanisms, this approach outperformed state-of-the-art deep learning models on real-world stock market data, demonstrating superior accuracy and profitability in trading simulations.

Graph-based methods present captivating paths to improve stock price forecasting, particularly when paired with symbolic dynamics and powerful feature extraction techniques. These techniques capture complex inter-dependencies and changing market structures that are difficult for traditional forecasting tools to handle by converting time series data into network representations. These techniques offer a more comprehensive understanding of financial markets, whether via community detection techniques, specific graph embeddings, transition sequences, or probabilities. It is anticipated that further research combining symbolic analysis and graph theory will lead to deeper understanding of the fundamental dynamics influencing market behavior as well as increased predictive accuracy, which will ultimately help the financial industry make better decisions.

Unlike traditional deep learning approaches that operate on Euclidean data, graph neural networks (GNNs) are well suited for capturing non-Euclidean relationships, such as co-movements of stock prices, sectoral dependencies, and investor behavior patterns. Cheng et al. [46] proposed a multi-modality graph neural network for financial time series forecasting, incorporating historical prices, media news, and knowledge graph-based relational data to model lead–lag effects between equities. Their method outperformed state-of-the-art baselines, achieving a micro-F1-score of 0.4838, a macro-F1-score of 0.46, and a weighted-F1-score of 0.48. Xu et al. [47] proposed a hierarchical graph neural network to classify price-limit-hitting stocks into two types, those that close at the daily price limit (Type I) and those that do not (Type II), by modeling hierarchical market states from the stock, industry, and market levels. Their method outperformed ALSTM, GCN, and GAT, achieving at least a 3.54% improvement in classification accuracy, with an average return ratio increase of 18.57% in the Shanghai Stock Exchange and 8.75% in the Shenzhen Stock Exchange. Chen et al. [48] proposed a graph convolutional feature-based convolutional neural network (GC–CNN) for stock trend prediction, integrating both individual stock information and market-wide relationships through an improved graph convolutional network and a Dual-CNN. Experimental evaluations on six Chinese stocks demonstrate that the GC–CNN model outperforms several traditional and deep learning-based methods, yielding more stable and higher returns, with annualized returns exceeding 20% for most stocks tested. Shi et al. [49] developed an integrated GCN-LSTM model for stock price movement prediction, incorporating four types of knowledge-based graphs (industry, region, concept, and volatility) to extract stock embeddings via GCN before passing them into an LSTM for temporal modeling. Their experiments on major Chinese stock indexes demonstrated the effectiveness of the proposed approach, achieving a peak accuracy of 57.81%. Yin et al. [50] developed a high-frequency trading system using graph attention LSTM (GALSTM) to model stock correlations and predict prices. Their approach achieved 44.71% annualized returns by integrating a multi-dimensional Hawkes process with an attention-based LSTM for dynamic portfolio management.

While GNN-based methods show significant promise in capturing structural dependencies in financial markets, several challenges remain in fully leveraging their potential for stock price forecasting. One limitation is the need for well-defined and meaningful graph structures, as financial relationships are often dynamic, latent, and non-stationary. However, adjacency-based approaches offer a computationally efficient way to integrate graph structures into deep learning models while mitigating some of these challenges. By representing stock relationships through sparse adjacency matrices, these methods allow for efficient message passing and structured feature extraction without requiring the full graph to be explicitly stored in memory. Techniques such as approximate nearest neighbor graphs, thresholded correlation networks, and knowledge-driven adjacency matrix construction help define relevant connections while avoiding excessive computational overhead.

Adjacency matrix-based relation extraction approaches have utilized several studies highlighting the importance of network analysis in stock market data. Khoojinea and Han [51] proposed a Stock Price Network Autoregressive Model (SPNAR) to analyze the behavior of the Chinese stock market during the 2015–2016 turbulence period by constructing an adjacency matrix-based financial network. Their approach defined stock relationships using an information-theoretic distance threshold to build sparse, interpretable adjacency matrices, which allowed for efficient estimation of market dependencies while maintaining computational feasibility compared to traditional vector autoregressive models. Zhao et al. [52] utilized a temporal network framework to analyze stock market evolution, emphasizing the role of adjacency matrices in capturing the dynamic dependencies among stocks. By constructing a supra-adjacency matrix that integrated time-evolving correlation-based networks, their approach preserved crucial temporal structures, allowing for improved portfolio selection and risk management, particularly through the identification of peripheral stocks.

In summary, symbolic dynamics and graph-based methods enable the extraction of structured patterns from financial time series, enhancing interpretability and predictive accuracy. Adjacency matrix-based techniques further optimize these approaches by efficiently capturing dynamic dependencies, facilitating scalable and robust financial modeling.

3 Symbolic dynamics for nonlinear financial time series analysis

Symbolic dynamics offers a powerful framework for analyzing complex time series by converting continuous-valued data into sequences of discrete symbols, thereby emphasizing qualitative patterns rather than precise numerical values. This transformation helps researchers capture essential structural and dynamical characteristics hidden in the data, making it especially suitable for studying inherently volatile and nonlinear domains like financial markets.

In the subsections that follow, the theoretical underpinnings of symbolic dynamics and their relevance to financial time series are first discussed, illustrating how these concepts can uncover meaningful patterns in non-stationary and chaotic market environments. The procedures for normalizing and discretizing continuous price signals into symbolic states are then described, with particular emphasis on how the choice of alphabet size and discretization methodology can influence both the resolution and transparency of the extracted patterns. Finally, the construction and analysis of state transition matrices from these symbolic sequences are introduced, demonstrating how graph-theoretic measures can be applied to represent and scrutinize the resulting networks, and utilized as inputs for deep learning models.

3.1 Exploration of temporal trends in stock price data

Financial time series, such as stock prices and trading volumes, are inherently noisy and exhibit highly non-stationary, nonlinear behavior, making traditional analytical methods challenging. Such time series data are characterized by persistent trends, cyclical fluctuations, and sudden regime shifts, often driven by external factors such as macroeconomic events, policy changes, or market sentiment.

Understanding the dynamics of stock price movements requires the decomposition of their behavior into clear, transparent temporal components. Trends, which signify sustained upward or downward movements, often reflect the long-term growth trajectories or structural declines of a company or sector. Cycles, on the other hand, capture recurring, periodic patterns over a defined period, such as quarterly earnings announcements or seasonal fluctuations. The presence of abrupt changes, such as price jumps or crashes, is typically attributed to unforeseen, high-impact shocks like geopolitical crises or earnings surprises.

One of the key challenges in analyzing financial data is distinguishing between meaningful trends and random fluctuations. Financial time series are often modeled as a combination of deterministic components and stochastic noise. The stochastic noise arises from factors such as market microstructure effects, investor sentiment, and high-frequency trading, which introduce short-term volatility and transient anomalies that may obscure the underlying patterns.

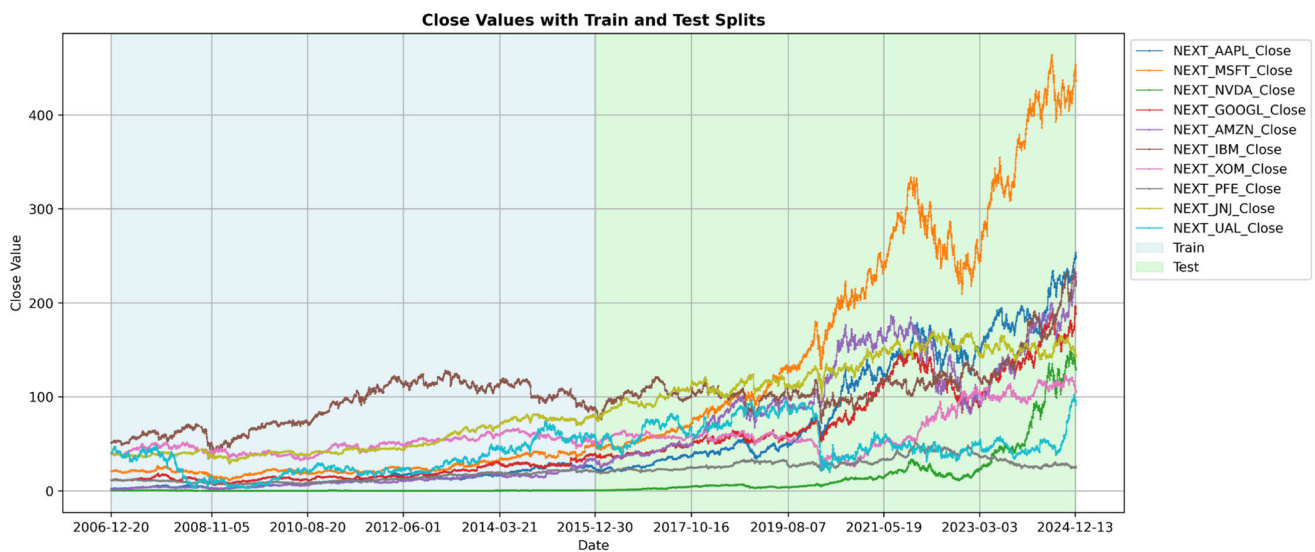
The dataset used in this study, sourced from Yahoo Finance, encompasses a diverse set of financial assets spanning multiple sectors and types, as summarized in Table 1. By drawing on publicly available historical data, these time series reflect real-market conditions and incorporate complexities inherent to financial environments. The primary focus is on corporate stocks, with target variables derived from the Close prices of assets in the Technology (e.g., AAPL, MSFT, NVDA), Healthcare (e.g., PFE, JNJ), Energy (e.g., XOM), Consumer Discretionary (e.g., AMZN), and Airlines (e.g., UAL) sectors. These target variables correspond to prediction horizons (e.g., 1-day, 10-day, 100-day) and are calculated as the percentage changes in future Close values relative to their current levels.

In addition to the Close prices, other key features used in the analysis include Volume (to capture trading activity) and High-Low Difference (to reflect daily price volatility). The dataset also incorporates data from non-target assets, including financial stocks (e.g., JPM, GS), commodities (e.g., GC = F for gold, CL = F for crude oil), and the EUR/USD exchange rate. These non-target assets provide additional contextual information for capturing broader market trends, improving the predictive power of the models.

Figure 1 illustrates the temporal evolution of stock prices across multiple corporate assets, covering a multi-year period and highlighting the segmentation of the dataset into training and testing sets, with a 50–50 split. In

Table 1 Financial Assets Utilized in this Study

Asset	Sector/Type	Used as Target
AAPL	Technology	Yes
MSFT	Technology	Yes
NVDA	Technology	Yes
GOOGL	Technology	Yes
AMZN	Consumer Discretionary	Yes
IBM	Technology	Yes
XOM	Energy	Yes
PFE	Healthcare	Yes
JNJ	Healthcare	Yes
UAL	Airlines	Yes
JPM	Financials	No
GS	Financials	No
GC = F	Commodity (Precious Metal)	No
SI = F	Commodity (Precious Metal)	No
HG = F	Commodity (Industrial Metal)	No
CL = F	Commodity (Energy)	No
NG = F	Commodity (Energy)	No
ZW = F	Commodity (Agriculture)	No
EUR/USD	Currency	No

**Fig. 1** Temporal Evolution of Close Values for Selected Stocks with Train and Test Splits Highlighted

this study, the chosen timeframe ensures that the training phase captures a historically rich and relatively stable market environment, while the test phase encompasses a variety of emerging trends and increased volatility, including the aftermath of global economic disruptions. Unlike traditional machine learning tasks where cross-validation is a widely accepted evaluation technique, its direct applicability to stock price forecasting is limited due to the strong temporal dependencies and the non-stationary nature of financial markets.

Standard k-fold cross-validation, commonly used in machine learning, assumes that data points are independent and identically distributed, an assumption that is fundamentally violated in time series forecasting. When applied to financial time series, k-fold cross-validation randomly partitions the dataset into folds, training on some subsets while validating on others [53, 54]. This introduces an unrealistic setting where later periods in time may

appear in the training data before earlier periods, leading to data leakage and inflated performance metrics that do not generalize to real-world forecasting tasks.

An alternative approach, rolling-origin cross-validation (also known as expanding window validation), maintains temporal consistency by incrementally expanding the training set while progressively validating on subsequent periods [18]. However, this method often results in models trained on earlier historical data, which may not fully capture the evolving market structures observed in the latter portions of the dataset. Moreover, rolling-origin validation tends to overweight earlier periods, which may not be representative of the structural changes and regime shifts occurring in financial markets.

Another common approach, sliding window validation, mitigates this issue by maintaining a fixed-length training window that shifts forward over time [18]. While this ensures that the model is continuously trained on the most recent data, it discards older information entirely, potentially losing valuable long-term dependencies crucial for robust market forecasting. Additionally, models trained using sliding window validation may struggle with abrupt market shifts, as the limited lookback period constrains their ability to recognize long-range trends.

To address these challenges, a simple but effective solution is the 50–50 temporal split adopted in this study. This division ensures that the model is trained on the initial half of the data, capturing historical trends and patterns, while the remaining half serves as an out-of-sample test set to evaluate predictive performance. By deliberately withholding more recent data until the testing stage, realistic forecasting conditions are simulated in which future market states are inherently uncertain and must be extrapolated from previously observed behaviors. This temporally consistent partitioning strategy enhances the robustness of performance evaluation, ensuring that the model is assessed under conditions that reflect its practical deployment in live trading scenarios. Furthermore, the equal division of the dataset avoids potential biases that arise when the training period is disproportionately larger or smaller than the test period, ensuring a balanced assessment of predictive stability across different market regimes.

The training phase, marked by the light blue region, provides the model with sufficient data to learn long-term sectoral trends, inter-market dependencies, and the evolution of cross-asset correlations. The testing phase, represented by the green region, covers more recent data and includes significant market fluctuations, such as the sharp price movements observed in the post-2020 period. This phase allows for an evaluation of the model's adaptability to regime shifts, external shocks, and structural breaks in financial markets. This period encapsulates several distinct market regimes, from initial recovery phases to periods of heightened investor uncertainty, spanning high-volatility events, macroeconomic disruptions, and changes in monetary policy frameworks, reflecting the shifting risk profiles and sentiment-driven behaviors characteristic of modern financial markets. By structuring the dataset to include both relatively stable and highly volatile conditions, the robustness of the model in varying market environments can be systematically assessed.

The Close values shown for each asset reflect the diversity of behaviors across sectors. Technology stocks, such as AAPL, MSFT, and NVDA, exhibit steady growth over time, punctuated by sharp upward trends, often synchronized with major product releases, breakthroughs in semiconductor technology, and shifts in investor sentiment toward high-growth assets. These price movements align with broader macroeconomic tailwinds, including increased digitalization and expansionary fiscal policies favoring innovation-driven sectors. Healthcare stocks, like PFE and JNJ, maintain more stable trajectories, with occasional volatility linked to industry-specific developments such as regulatory approvals, patent expirations, mergers and acquisitions, or global health crises that reshape demand for pharmaceutical and biotech products. Energy sector stocks, such as XOM, are influenced by commodity price fluctuations, which are, in turn, driven by geopolitical tensions, supply chain disruptions, and changes in global demand for crude oil and natural gas. These fluctuations introduce periodic boom-and-bust cycles, with price reversals occurring in response to macroeconomic indicators such as inflationary pressures and shifts in energy policy. These observed patterns reinforce that stock price dynamics are not uniform but strongly influenced by the intrinsic characteristics and external pressures unique to each sector. The heterogeneity of these patterns underscores the necessity of a modeling approach that captures both short-term dependencies and long-term structural shifts, making sector-aware forecasting frameworks particularly valuable.

To further ensure that the dataset division captures a realistic market representation, the selection of the training and testing periods was carefully designed to encompass diverse economic conditions. The training phase spans a period characterized by relative market stability, allowing the model to learn fundamental sectoral trends, cross-asset dependencies, and macroeconomic influences that define long-term financial dynamics. Meanwhile, the testing phase is deliberately structured to include time frames of heightened market turbulence, such as the COVID-19 pandemic and post-pandemic recovery, ensuring that the model's predictive capability is evaluated under extreme volatility and regime shifts.

This diverse economic landscape necessitates a modeling approach that not only learns from historical patterns but also accounts for the broader market context in which financial assets operate. To comprehensively capture the interconnected nature of financial markets, the modeling framework does not solely rely on the historical behavior of each individual target asset in isolation. Instead, for each target asset, features derived from all other target assets, as well as relevant non-target assets, are incorporated into the predictive process. This design choice acknowledges the interdependence of financial instruments, where price movements and volatility in one sector may propagate across others due to macroeconomic factors, investor sentiment shifts, or structural market relationships. By integrating sector-wide information, the model leverages cross-asset correlations to refine its forecasts, improving robustness against unpredictable market fluctuations. Additionally, the inclusion of non-target assets such as financial stocks and commodities enhances contextual awareness by incorporating signals that reflect broader market conditions.

3.2 Symbolization and state transition graph constructing methodology

Symbolization, a cornerstone of symbolic dynamics, transforms continuous-valued time series data into sequences of discrete symbols, thereby highlighting dynamic patterns and reducing complexity to a manageable, interpretable form. The implementation analyzes segments of length 100 timesteps, considering each window from time t down to $t - 99$, forwarding the sectioning window by 1 timestep at each iteration. This approach ensures that each symbolic representation and subsequent analysis is based on a consistent temporal context of 100 consecutive observations. By maintaining a fixed window size, the framework ensures stationarity within local segments while preserving the sequential dependencies necessary for capturing market dynamics.

The analyze begins with the scaling the all time series first, then normalization of the windowed time series performed, ensuring scale invariance and facilitating uniform analysis across multiple datasets. This is achieved by mapping the series x_t to the interval $[0,1]$ using:

$$x_t^{\text{normalized}} = \frac{x_t - \min(x)}{\max(x) - \min(x)} \quad (1)$$

Following normalization, the signal is partitioned into a finite number of amplitude states, defined by an alphabet of size n . These amplitude states $\{a_0, a_1, \dots, a_{n-1}\}$ are determined by dividing the normalized range $[0,1]$ into equal intervals for $n = 10$. This symbolic representation retains critical information about the signal's magnitude while discarding finer numerical details. The discretization ensures a compact representation of the signal while balancing the trade-off between resolution and computational efficiency. Let $\{a_0, a_1, \dots, a_{n-1}\}$ represent a finite alphabet of amplitude states, where each state corresponds to a uniform partitioning of the interval $[0,1]$:

$$a_i = \left[\frac{i}{n}, \frac{i+1}{n} \right), i = 0, 1, \dots, n-1 \quad (2)$$

For each x_t^{norm} , the amplitude symbol is assigned as:

$$A_t = a_i \text{ if } x_t^{\text{norm}} \in a_i \quad (3)$$

To capture directional changes, a trend state, T_t , is assigned based on the first-order difference:

$$\Delta x_t = x_t - x_{t-1} \quad (4)$$

The trend state, slope, is then categorized into three discrete states:

$$T_t = \begin{cases} d, & \text{if } \Delta x_t < 0 \text{ (Decreasing)} \\ s, & \text{if } \Delta x_t = 0 \text{ (stable)} \\ i, & \text{if } \Delta x_t > 0 \text{ (Increasing)} \end{cases} \quad (5)$$

The combination of amplitude and trend states allows for the characterization of not only magnitude-based fluctuations but also the directional evolution of the financial time series, providing a richer representation of the underlying dynamics. By combining the amplitude and trend states, each timestep is assigned a composite symbolic representation:

$$S_t = (A_t, T_t) \quad (6)$$

where S_t results in a sequence of symbolic states that describe both magnitude and directional dynamics.

Given the sequence of symbolic states $\{S_t\}_{t=1}^N$, the transition matrix $T \in \mathbb{R}^{n \times n}$ is constructed that quantifies the transitions between symbolic states and is defined as for $m = n \times 3$ total number of unique symbolic states (amplitude-slope combinations). Each element in the matrix is computed as:

$$T_{ij} = \frac{\sum_{t=1}^{N-1} 1(S_t = i, S_{t+1} = j)}{\sum_{t=1}^{N-1} 1(S_t = i)} \quad (7)$$

where $1(\cdot)$ is an indicator function that returns 1 if the condition is met and 0 otherwise. This ensures that:

$$\sum_{j=1}^m T_{ij} = 1, \forall i \quad (8)$$

confirming that each row of T represents a valid probability distribution.

The state transition matrix can be visualized as a graph, $G = (V, E)$, where V consists of m nodes, each corresponding to a unique state, and E consists of directed edges weighted by the transition counts for this purpose. This approach reveals temporal structures and facilitates graph-theoretic analysis of complexity and cyclic behavior. Figure 2 illustrates this using Apple Inc.'s stock price data: Part (a) shows the normalized time series with symbolic states representing discretized amplitude and trends, while part (b) depicts the state transition graph, with edge weights showing number of transitions between those nodes.

The transition matrix is mathematically analogous to a Markov transition matrix, capturing first-order dependencies in the symbolic sequence. This methodology facilitates the comparison of time series across assets and timeframes, providing a robust framework for understanding the qualitative dynamics of financial markets. By encoding both amplitude and directional information, the symbolic representation emphasizes significant qualitative features while reducing noise and preserving essential characteristics of the signal.

To rigorously assess whether the transition matrices satisfy the Markov property, several statistical tests grounded in matrix analysis, information theory, and statistical inference were employed. The Markov property

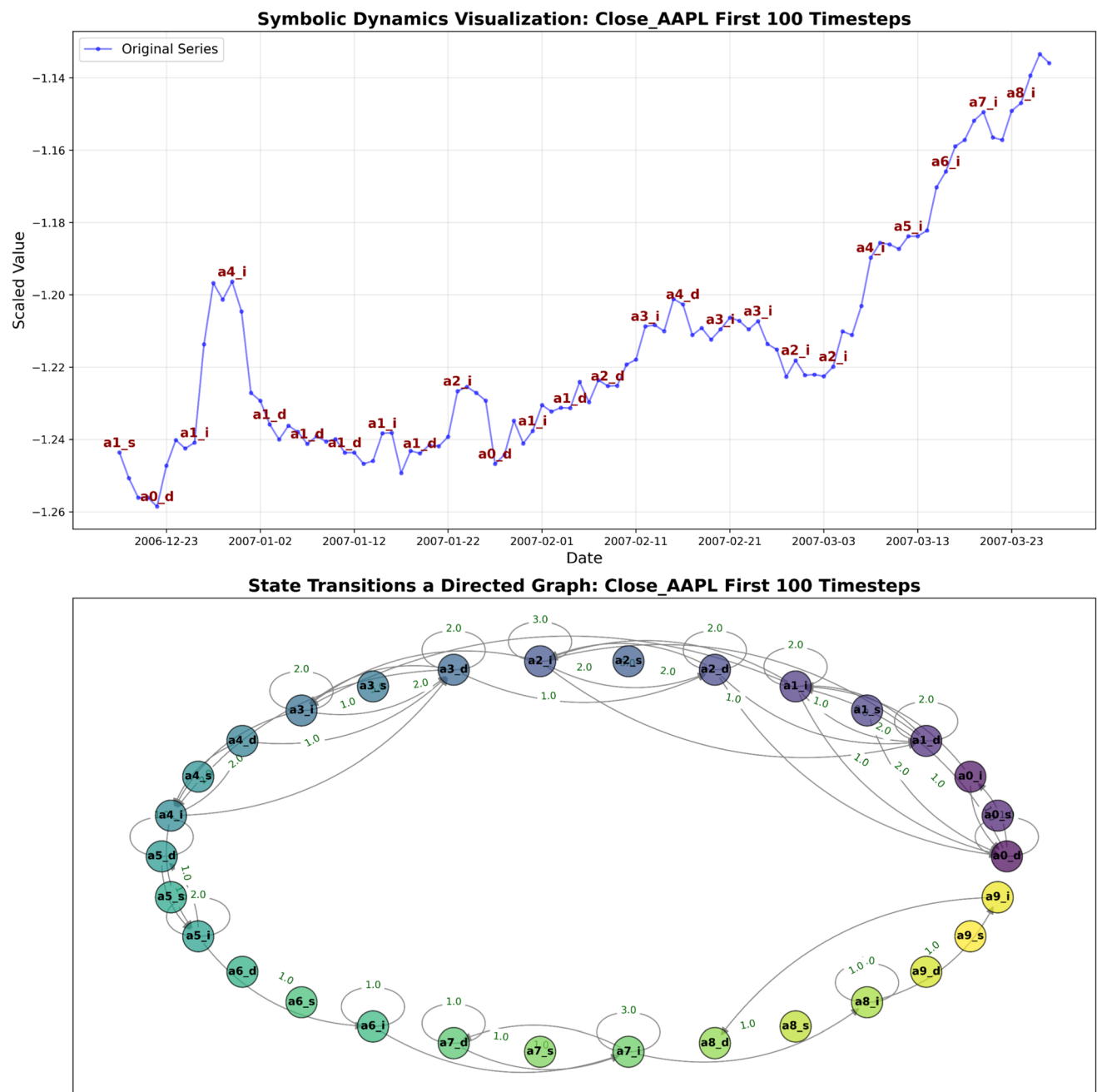


Fig. 2 **a** Symbolic Dynamics Visualization of AAPL Close Values for the First 100 Timesteps, Highlighting Assigned States; **b** Corresponding State Transition Graph Representing the Temporal Dynamics of Symbolic States

states that a stochastic process X_t is Markovian if the probability of transitioning to the next state depends only on the present state, and not on the sequence of past states:

$$P(X_{t+1}|X_t, X_{t-1}, \dots, X_0) = P(X_{t+1}|X_t) \quad (9)$$

To examine this assumption, the empirical second-step transition matrix, $P_{\text{empirical}}^{(2)}$, is computed and compared to the theoretical second-step transition matrix derived from the Chapman–Kolmogorov equation [55–57]:

$$P_{\text{theoretical}}^{(2)} = P^1 \bullet P^1 = P^2 \quad (10)$$

where P^1 is the one-step transition matrix. The following statistical measures were used to evaluate the discrepancy between these matrices:

- **Frobenius Norm Difference:** The Frobenius norm measures the Euclidean difference between the empirical and theoretical second-step transition matrices [58, 59]:

$$d_F = \|P_{\text{empirical}}^{(2)} - P_{\text{theoretical}}^{(2)}\|_F \quad (11)$$

where the Frobenius norm is defined as:

$$\|A\|_F = \sqrt{\sum_{i,j} |a_{ij}|^2} \quad (12)$$

A low Frobenius norm indicates that the observed transition structure closely follows the Markovian assumption.

- **Kullback–Leibler (KL) Divergence:** The KL divergence quantifies how much information is lost when using the theoretical transition matrix instead of the empirical one [60, 61]:

$$D_{KL}(P_{\text{empirical}}^{(2)} \| P_{\text{theoretical}}^{(2)}) = \sum_{i,j} P_{\text{empirical}}^{(2)}(i,j) \log \frac{P_{\text{empirical}}^{(2)}(i,j)}{P_{\text{theoretical}}^{(2)}(i,j)} \quad (13)$$

A low KL divergence suggests that the empirical transition probabilities follow the theoretical Markovian expectations.

- **Pearson Correlation Coefficient:** The Pearson correlation coefficient measures the linear relationship between the flattened empirical and theoretical transition matrices [62, 63]:

$$\rho(P_{\text{empirical}}^{(2)}, P_{\text{theoretical}}^{(2)}) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2} \sqrt{\sum (Y_i - \bar{Y})^2}} \quad (14)$$

where X and Y are vectorized forms of the empirical and theoretical matrices, respectively. A high correlation indicates strong agreement between observed and expected Markovian transitions.

- **Likelihood Ratio Test (LRT):** To formally test the first-order Markov assumption against a second-order model, which can be compared in terms of the log-likelihoods of the first-order transition model and the second-order empirical transition model [64–66]:

$$\lambda = -2 \left(\log L(P^{(1)}) - \log L(P^{(2)}) \right) \quad (15)$$

where $L(P)$ is the likelihood of observing the given transitions under the respective model. Under the null hypothesis of first-order Markovianity, λ follows a chi-square distribution [67, 68]:

$$\lambda \sim \chi^2(k) \quad (16)$$

with k degrees of freedom determined by the number of free parameters in the transition matrices. A statistically significant likelihood ratio suggests that higher-order dependencies may be present.

These tests collectively provide a rigorous mathematical framework to assess the validity of the first-order Markov assumption in the financial time series. This multi-faceted evaluation captures local deviations in transition probabilities and broader structural inconsistencies in state dynamics by leveraging complementary statistical and information-theoretic measures. Significant deviations from the expected Markov structure may indicate hidden dependencies, long-range correlations, or structural non-stationarities, suggesting that additional higher-order memory effects or regime shifts influence market behavior. Such deviations could emerge from latent exogenous factors, behavioral biases, or autocorrelated volatility patterns that are not fully captured within the first-order Markovian framework.

Figure 3 presents the cumulative distributions of four Markovianity test metrics—Frobenius norm difference, KL divergence, Pearson correlation, and likelihood ratio—comparing the train and test periods to assess the stability of first-order dependencies in transition matrices. These results provide empirical evidence on whether transition structures remain consistent over time or undergo structural breaks. The findings from these tests contribute to a deeper understanding of how well symbolic-state-based models capture the underlying probabilistic structure of financial markets.

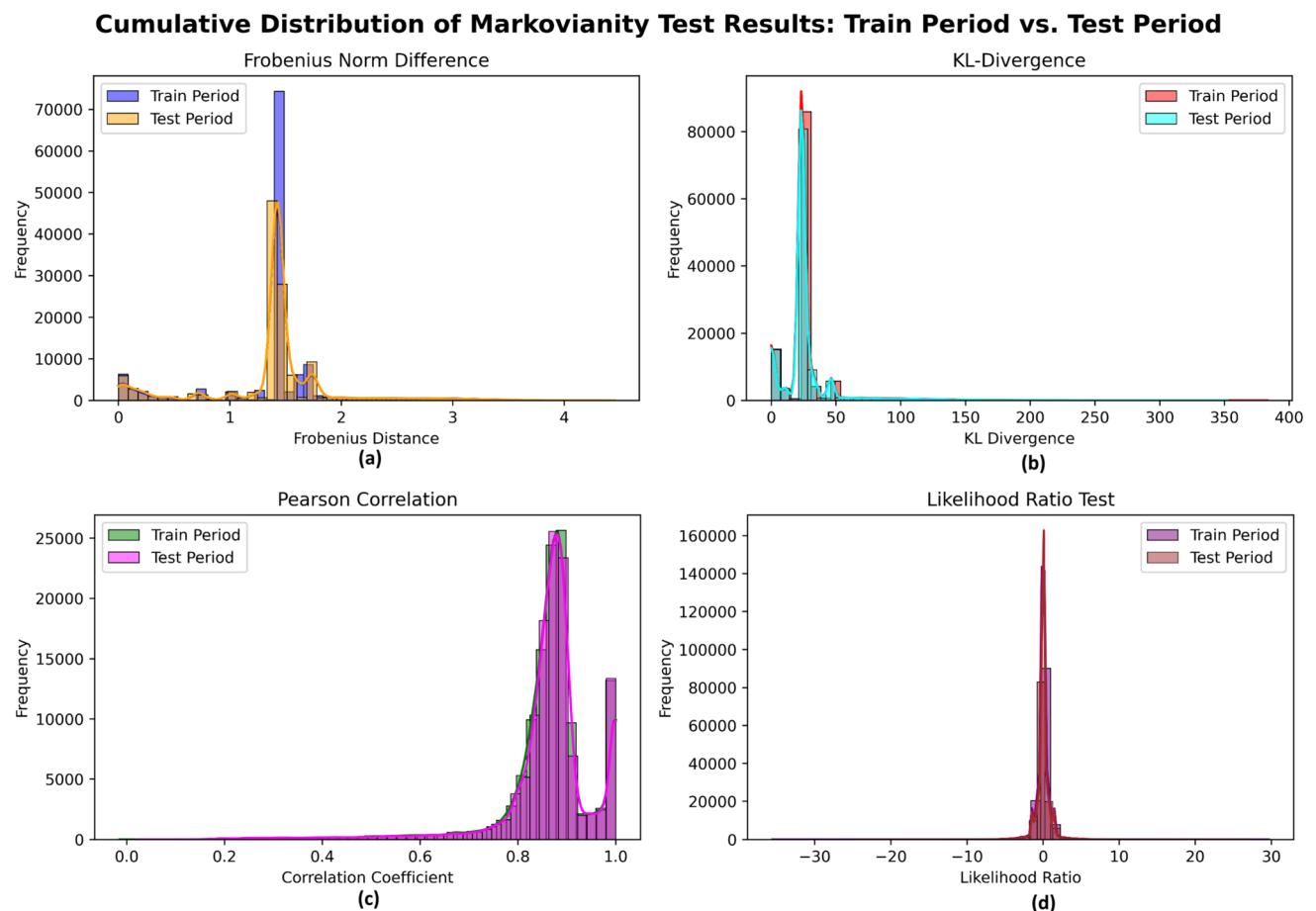


Fig. 3 Cumulative Distribution of Markovianity Tests; **a** Frobenius Norm Difference; **b** KL Divergence; **c** Pearson Correlation; and **d** Likelihood Ratio

The cumulative distributions of the Markovianity test results provide insights into the validity of the first-order Markov assumption across training and test periods. The Frobenius norm difference distribution is around 1.5, indicating a systematic deviation between empirical and theoretical second-step transition matrices. Both periods exhibit similar distributions, suggesting that the transition structure remains relatively stable, with no abrupt shifts in dependency patterns. Minor variations between the train and test periods suggest some non-stationarity but do not indicate a fundamental change in the underlying Markovian behavior.

The KL divergence distributions exhibit a sharp peak near zero, with a long-tailed spread. This indicates that while many transition matrices remain close to their theoretical counterparts, some cases deviate, especially in the test period. The extended tail in the test period suggests increased uncertainty or structural shifts in the temporal dependencies of the financial time series. This finding implies that while first-order dependencies are dominant, occasional deviations from strict Markovianity occur.

The Pearson correlation values are predominantly near 0.85–1.0, reflecting a strong agreement between empirical and theoretical transition matrices. This supports the assumption that the first-order Markov model captures short-term dependencies well in most cases. However, the slight spread in the test period suggests increased variability in temporal dynamics, indicating that certain dataset segments may exhibit higher-order dependencies. Despite this, the overall consistency between periods supports the robustness of the Markov-based modeling approach.

The likelihood ratio test results are heavily centered around zero, indicating that the first-order Markov assumption is often not firmly rejected. The similarity between train and test period distributions suggests that dependencies between states remain stable over time, although minor fluctuations may signal evolving market conditions. These findings provide evidence that symbolic dynamics-based transition matrices effectively capture short-term dependencies.

Although financial time series are modeled as transition graphs and corresponding transition matrices, their representation within neural network architectures requires a structured temporal encoding. To achieve this, transition matrices are stacked with a temporal dimension, where each sample consists of a sequence of five consecutive matrices, incorporating the previous four matrices as observations for the current step. This approach allows the network to capture temporal dependencies beyond a single-step transition, effectively embedding historical dynamics into the learning process. By structuring the input in this way, the neural network can exploit both short-term Markovian transitions and potential deviations from strict first-order assumptions.

4 Convolutional recurrent neural network and forecasting strategy

Forecasting time series data, particularly in the context of financial markets, requires a methodology that can handle complex temporal dependencies, non-stationary, and multivariate interactions. This section details the end-to-end pipeline, highlighting how symbolic transformations, tensor construction, and hybrid neural architectures are combined to provide a robust and scalable forecasting strategy. Figure 4 illustrates the complete pipeline for transforming time series data into structured inputs for a CRNN and the corresponding architecture used for predictive modeling. This pipeline encompasses four key stages: symbolic dynamics-based preprocessing, transition matrix calculation and normalization, temporal stacking into tensors, and the design of a neural network that integrates convolutional and recurrent layers for robust time series forecasting.

4.1 Construction temporally stacked transition matrices as tensors

To extend the temporal dynamics captured by state transition matrices, temporally stacked tensors are constructed that aggregate sequential information over multiple time windows. This approach enables recurrent neural network (RNN)-based models to process transition matrices in a temporally structured format, capturing dependencies not just within individual windows but also across successive windows. Given that transition matrices T

Input Tensor Genration Process and Convolutional Recurrent Neural Network Architecture

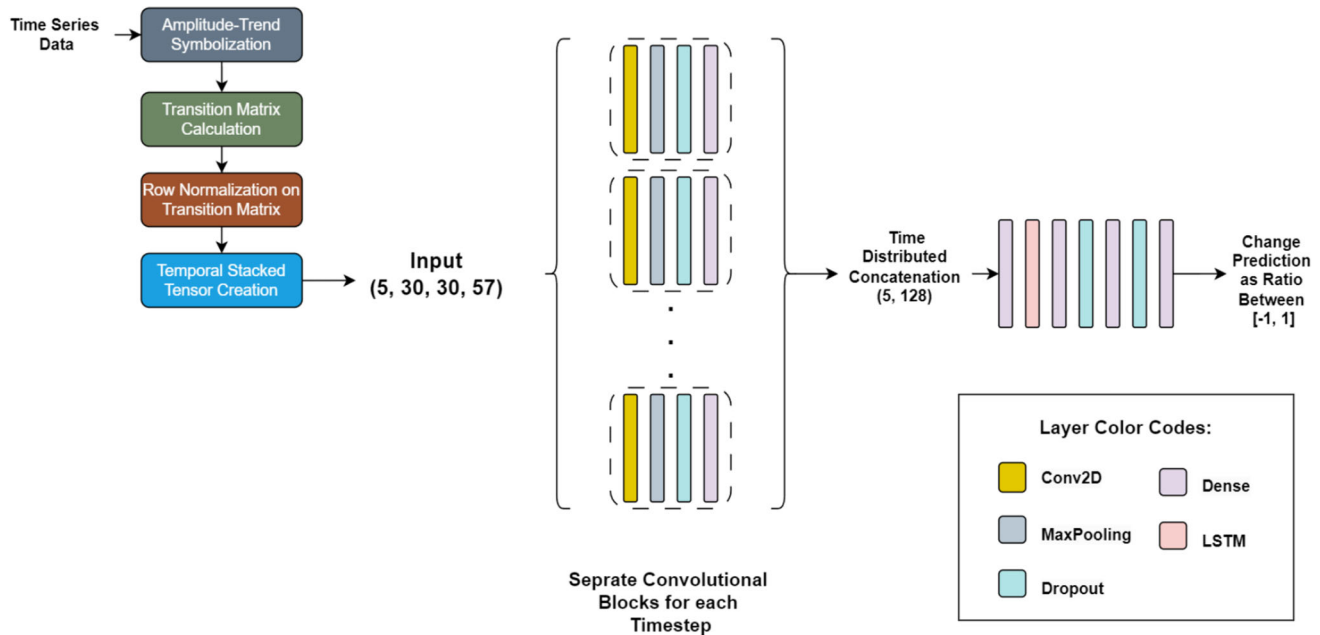


Fig. 4 Overview of Input Tensor Generation and Convolutional Recurrent Neural Network Architecture

are computed for each timestep t over the interval $[t, t-99]$, the input tensors incorporate transition matrices from multiple consecutive steps, specifically $t-4$ to t for each transition matrix in timestep t , forming a temporally structured sequence. This approach enables recurrent neural network (RNN)-based models to process transition matrices in a time-aware manner, capturing dependencies not just within individual observation windows but also across successive transitions.

Each stacked tensor X_{tensor}^t is thus defined as:

$$X_{\text{tensor}}^t = [T_{t-4}, T_{t-3}, T_{t-2}, T_{t-1}, T_t] \quad (17)$$

The entire tensor X_{tensor}^t encapsulates the sequential evolution of transition probabilities across recent market states, which have the shape of:

$$X_{\text{tensor}}^t \in \mathbb{R}^{(5,m,m,f)} \quad (18)$$

where

- 5 is the temporal dimension, representing the stacked tensors,
- m is the number of symbolic states (e.g., 30 symbolic states in the discretization process), and
- f is the number of time series features utilized, which is 57 coming from 19 assets.

The temporally stacked tensor encodes both short-term and long-term dynamics, making it an ideal input for RNN-based models, such as long short-term memory (LSTM) neural networks [54, 69, 70]. The inclusion of multiple temporal windows enables these models to leverage sequential patterns across time, enhancing their ability to predict future dynamics. Specifically, the CRNN architecture utilizes convolutional layers to capture spatial correlations within the transition matrices, followed by recurrent layers that process the temporal

dependencies encoded in the stacked tensor. This hybrid architecture ensures that both the intra-window and inter-window patterns are effectively captured, providing a comprehensive understanding of the temporal dynamics underlying the time series.

Furthermore, the temporal stacking framework aligns with the concept of capturing hierarchical temporal structures, where shorter temporal windows capture high-frequency transitions and longer sequences allow the model to learn low-frequency trends. This hierarchical processing facilitates robust forecasting, particularly in scenarios where financial time series exhibit complex, multi-scale temporal dependencies.

4.2 Model architecture and the role of convolutional layers

The proposed CRNN architecture is designed to effectively process temporally stacked tensors that encode symbolic-state transitions across consecutive time windows. In our approach, each temporal segment is represented by a series of transition matrices that reflect evolving state dynamics over time, providing a structured view of the underlying temporal patterns. Each tensor slice consists of 5 transition matrices, corresponding to consecutive segments of the time series, ensuring that the network receives a coherent block of historical information at each step. A notable characteristic of these transition matrices is their inherent sparsity, as many transitions between states may not occur frequently, leading to zero or near-zero entries. This sparsity presents a unique computational and modeling challenge that necessitates specialized preprocessing layers.

To address this, convolutional layers are applied to each transition matrix individually within the temporally stacked tensor. By processing each matrix separately, the network can isolate local spatial correlations without interference from adjacent matrices, ultimately reducing the risk of blending together unrelated patterns. This independent processing ensures that the spatial correlations within each matrix are preserved and allows the network to capture patterns specific to individual temporal windows. The convolutional blocks serve multiple purposes: They extract local spatial patterns from the transition matrices, reduce the impact of noise due to sparsity, and identify meaningful state transitions that contribute to the temporal dynamics. Each convolutional block is composed of Conv2D layers, followed by max-pooling layers to down sample and highlight the most prominent spatial features. By accentuating key structural dependencies and suppressing superfluous detail, these operations help create a more compact and informative representation of the data, laying a strong foundation for subsequent temporal modeling.

The processed features are subsequently passed to recurrent layers, specifically LSTM units, which are designed to model the temporal dependencies across the stacked windows [34, 54, 70, 71]. LSTMs excel at retaining and updating information over extended sequences, making them well suited for capturing the evolution of state transitions as the time series unfolds. This sequential modeling allows the network to capture both short-term and long-term dynamics in the time series, leveraging the rich spatial features extracted by the convolutional layers. In essence, the CRNN design combines the complementary strengths of convolutional and recurrent architectures, and convolutions handle the spatial intricacies of the transition matrices, while LSTMs contextualize these findings within a temporal narrative, resulting in a more robust and adaptive forecasting framework.

4.3 Target variables and final forecast predictions

In this modeling framework, the target variables represent the ratio of change between the current and future values of the Close prices, normalized to fall within the range $[-1, 1]$. This transformation ensures numerical stability during training and aligns the target variable with the output activation function of the neural network. The ratio of change for each asset is computed as:

$$y_t = \frac{\text{Close}_{t+h} - \text{Close}_t}{\text{Close}_t} \quad (19)$$

where y_t is the target variable at time t , Close_t is the current Close price, and Close_{t+h} is the Close price at a future horizon h . This representation captures the relative change in the asset price over the prediction horizon, enabling the model to focus on the proportional dynamics rather than absolute values.

During the inference phase, the model outputs predictions for y_t , the ratio of change, based on the processed temporal and spatial features from the input tensors. To derive the final forecasted Close price, these ratio predictions are transformed back into the original scale of the time series using the current Close price. The forecasted value is calculated as:

$$\widehat{\text{Close}}_{t+h} = \text{Close}_t \times (1 + \hat{y}_t) \quad (20)$$

where $\widehat{\text{Close}}_{t+h}$ is the forecasted future Close price and $\hat{y}_t$ is the model's predicted ratio of change.

This formulation ensures that the model predictions remain consistent with the original scale of the financial time series while preserving the coherence of the ratio predictions. By expressing the target variable as a ratio, the model can generalize across assets with different price ranges. This standardization ensures that the loss function treats all assets equally, avoiding bias toward high-value stocks. Ratio-based targets capture relative changes, making the model more robust to non-stationary behavior in the time series, such as price trends or volatility shifts.

5 Results and discussion

This section examines the outcomes and implications of the forecasting strategy, evaluating its effectiveness across three different forecast horizons. The forecasting results for 1-day, 10-day, and 100-day prediction horizons demonstrate the model's ability to adapt to varying temporal dynamics across different market conditions. As illustrated in Figs. 5, 6, 7, the predictions are analyzed within three distinct time intervals: pre-COVID, COVID, and post-COVID. Each interval exhibits unique patterns and levels of volatility and distinct market dynamics that influence investment strategies and risk assessment. The pre-COVID period is characterized by stable economic conditions, gradual trends, and predictable market behavior, providing a relatively low-risk environment where the model achieves high forecasting accuracy. The COVID period, in contrast, reflects extreme volatility and rapid shifts driven by global uncertainty, making it a highly challenging period for accurate predictions. This interval is crucial for robust forecasting models to support decisions in crisis scenarios, where market shocks demand rapid adaptations. Finally, the post-COVID period signifies a recovery phase with emerging stability, allowing for more reliable forecasting and strategic planning.

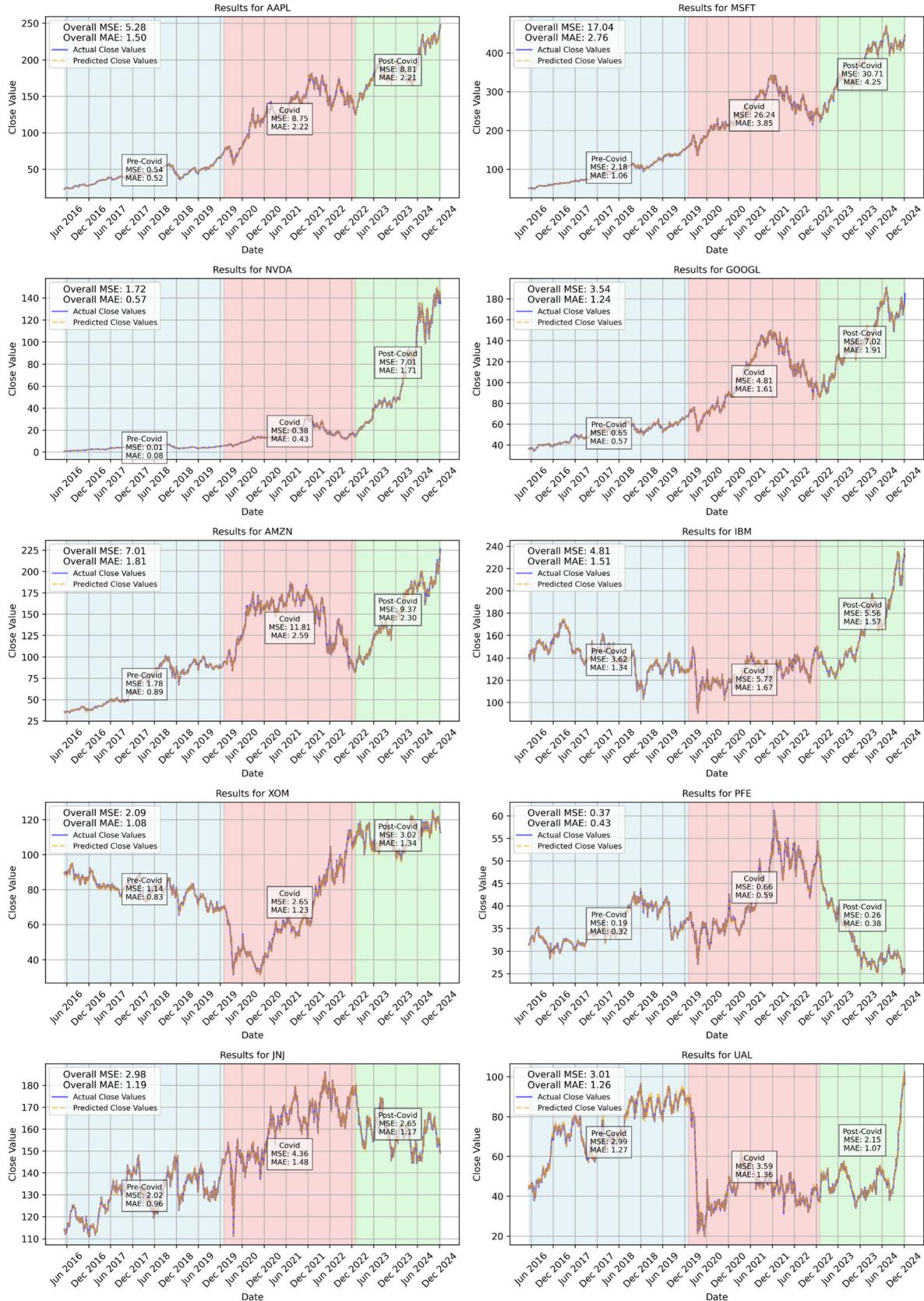
The model's performance is quantitatively assessed using two widely adopted metrics: mean absolute error (MAE) and mean squared error (MSE). These metrics provide complementary perspectives on the accuracy of predictions. Mathematically, MAE is defined as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (21)$$

where y_i represents the actual value, $\hat{y}_i$ is the predicted value, and n is the total number of predictions. MAE captures the average magnitude of prediction errors, treating all deviations equally regardless of their direction, making it intuitive for understanding the typical size of errors.

On the other hand, MSE is defined as:

Forecast Results for 1 Day Prediction Horizon



◀**Fig. 5** Forecast Results for 1-Day Prediction Horizon

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (22)$$

MSE takes the squared differences between actual and predicted values. By squaring the errors, MSE penalizes larger deviations more heavily than smaller ones, providing sensitivity to significant outliers in the predictions.

For shorter prediction horizons, such as the 1-day forecast, the model consistently achieves lower values for both MAE and MSE, reflecting a high degree of accuracy in short-term predictions. Notably, during the stable pre-COVID period, the reduced market volatility results in a lower variance of prediction errors, enabling the model to perform particularly well. This stability minimizes the influence of noise and confounding factors, leading to more reliable and precise predictions during this interval. The explicit focus on short-term dynamics and the low noise levels in pre-COVID data further enhance the performance metrics in this context.

During the COVID interval, characterized by heightened volatility and abrupt market fluctuations, the model's predictive performance becomes more asset- and sector-dependent. Technology stocks, such as AAPL and MSFT, exhibit higher errors due to the pronounced price jumps and sharp trends. In contrast, healthcare stocks like PFE and JNJ maintain relatively stable performance, highlighting sector-specific resilience even amid global uncertainty. Post-COVID, the model demonstrates improved predictive capability for most stocks, capturing recovery trends and sector-specific growth patterns, particularly for energy stocks like XOM, which benefit from cyclic rebounds in commodity markets.

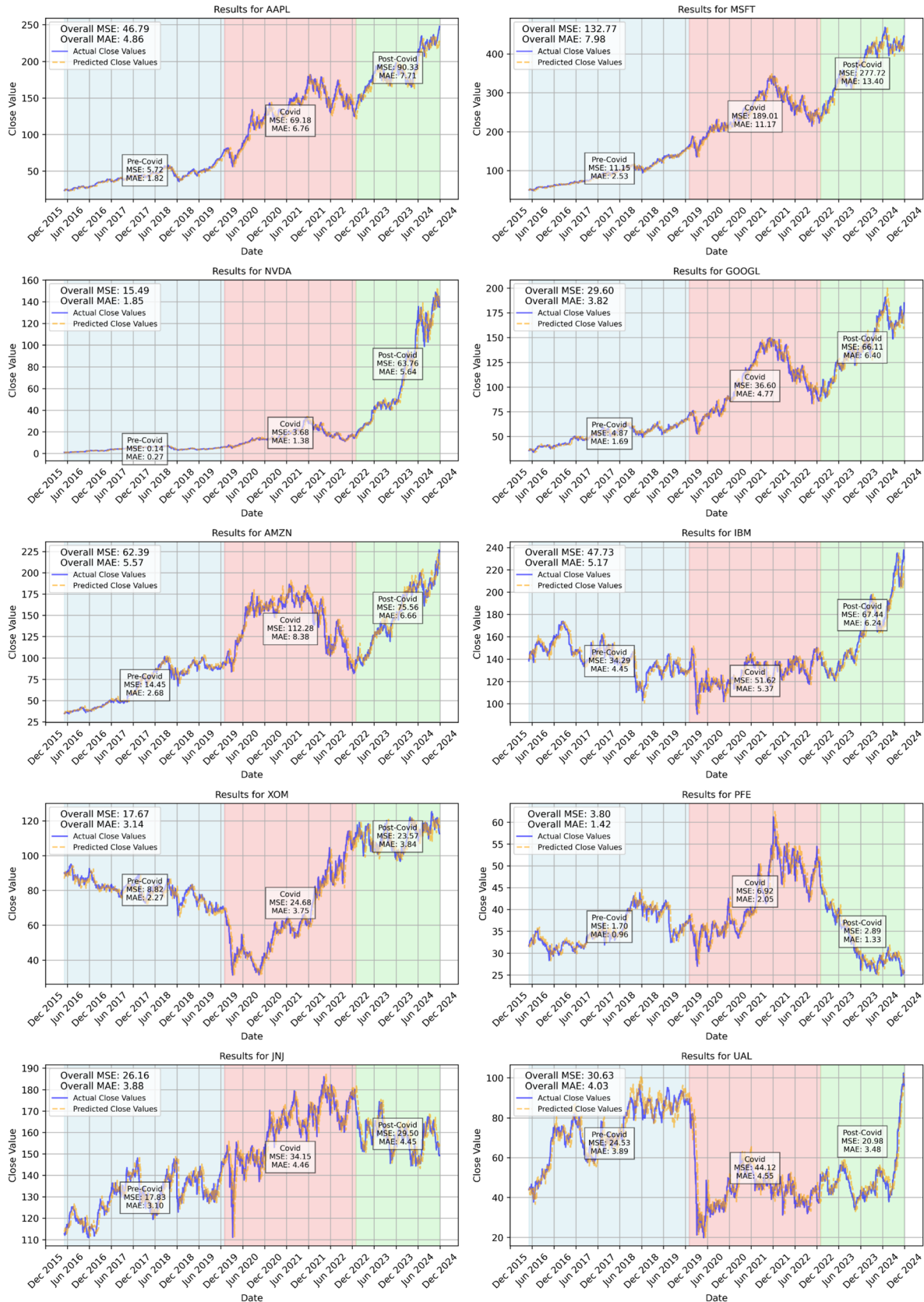
For longer horizons (e.g., 10-day and 100-day predictions), errors understandably rise as uncertainty compounds over time. However, the temporal stacking of transition matrices, coupled with the CRNN architecture, allows the model to retain meaningful sequential information, yielding competitive performance even during challenging periods. These results emphasize the importance of integrating symbolic dynamics and deep recurrent structures to capture the intricate, long-range dependencies inherent in financial time series. Longer-term horizons offer perspectives on more general market trends at the expense of greater uncertainty, while short-term horizons provide higher precision but may lack broader strategic insights.

The quantitative analysis of the forecasting results across pre-COVID, COVID, and post-COVID intervals reveals significant variations influenced by market conditions and prediction horizons. During the pre-COVID period, marked by relatively stable market dynamics, the model exhibited superior accuracy, achieving the lowest errors across all horizons. For instance, AAPL's 1-day horizon MSE and MAE were 0.54 and 0.52, respectively, while NVDA showed exceptional precision with an MSE of 0.01 and MAE of 0.08 for the same horizon. However, as the prediction horizon extended to 10 days and 100 days, the errors naturally increased due to the compounding uncertainties, though the pre-COVID errors remained significantly lower than in the volatile COVID period.

The COVID interval, characterized by heightened volatility and market disruptions, posed considerable challenges for the model. For instance, MSFT's MSE rose sharply to 26.24 for the 1-day horizon and 189.01 for the 10-day horizon, reflecting the difficulty in capturing abrupt market shifts. Longer horizons, such as the 100-day forecast, showed even more pronounced errors, with AMZN's MSE reaching 1489.84. Similarly, GOOGL experienced a surge in MSE from 4.87 in the pre-COVID period to 36.00 during COVID for the 10-day horizon. The increased errors highlight the need for robust modeling approaches to adapt to extreme market conditions.

In the post-COVID phase, while market conditions began stabilizing, residual volatility remained, leading to moderate improvements in forecasting accuracy compared to the COVID interval. For instance, XOM's MSE for the 1-day horizon reduced from 2.65 during COVID to 3.02 post-COVID, and AAPL's 100-day horizon MSE dropped from 642.19 during COVID to 604.76 post-COVID. However, the errors remained higher than in the

Forecast Results for 10 Days Prediction Horizon



◀**Fig. 6** Forecast Results for 10-Day Prediction Horizon

pre-COVID period, reflecting the ongoing challenges in modeling long-term forecasts in dynamic post-pandemic markets.

Table 2 presents forecasting metrics (MSE and MAE) for a range of assets—AAPL, MSFT, NVDA, GOOGL, AMZN, IBM, XOM, PFE, JNJ, and UAL—across three forecast horizons (1-day, 10-day, and 100-day) and three market regimes (pre-COVID, COVID, and post-COVID).

Figure 8 quantitatively evaluates the percentage changes in MSE and MAE across three prediction horizons (1 day, 10 days, and 100 days) for multiple assets during the transitions between pre-COVID, COVID, and post-COVID periods. As depicted in the figure, medium-term (10 days) and long-term (100 days) horizons demonstrate a persistent pattern of volatility; however, the percentage increases in MSE generally decline for most assets, reflecting reduced sensitivity to COVID-related disruptions over longer timeframes. Conversely, short-term predictions (1 Day) exhibit pronounced spikes in MSE, particularly during the transition from pre-COVID to COVID, indicating that these horizons are more vulnerable to abrupt market fluctuations induced by the pandemic.

The technology sector demonstrates pronounced increases in prediction errors, particularly during the transition from pre-COVID to COVID. For instance, NVDA shows extreme increases in MSE, reaching approximately 3500% for the 1-day horizon, indicating a significant challenge in capturing the volatility of technology stocks during the early pandemic phase. Similarly, MSFT exhibits a 1500% increase in 1-day MSE and a sustained rise of over 1000% for longer horizons (10 days and 100 days). AAPL and GOOGL, while exhibiting more moderate increases (below 1000%), still reflect elevated modeling challenges due to increased market uncertainty.

In contrast, IBM, as a more stable technology stock, shows relatively smaller error increases, with MSE changes remaining below 500% across all horizons. These findings suggest that technology sector stocks, particularly high-growth and volatile ones like NVDA and MSFT, posed significant modeling challenges during the pandemic, requiring adaptive strategies to manage sudden shifts in market dynamics.

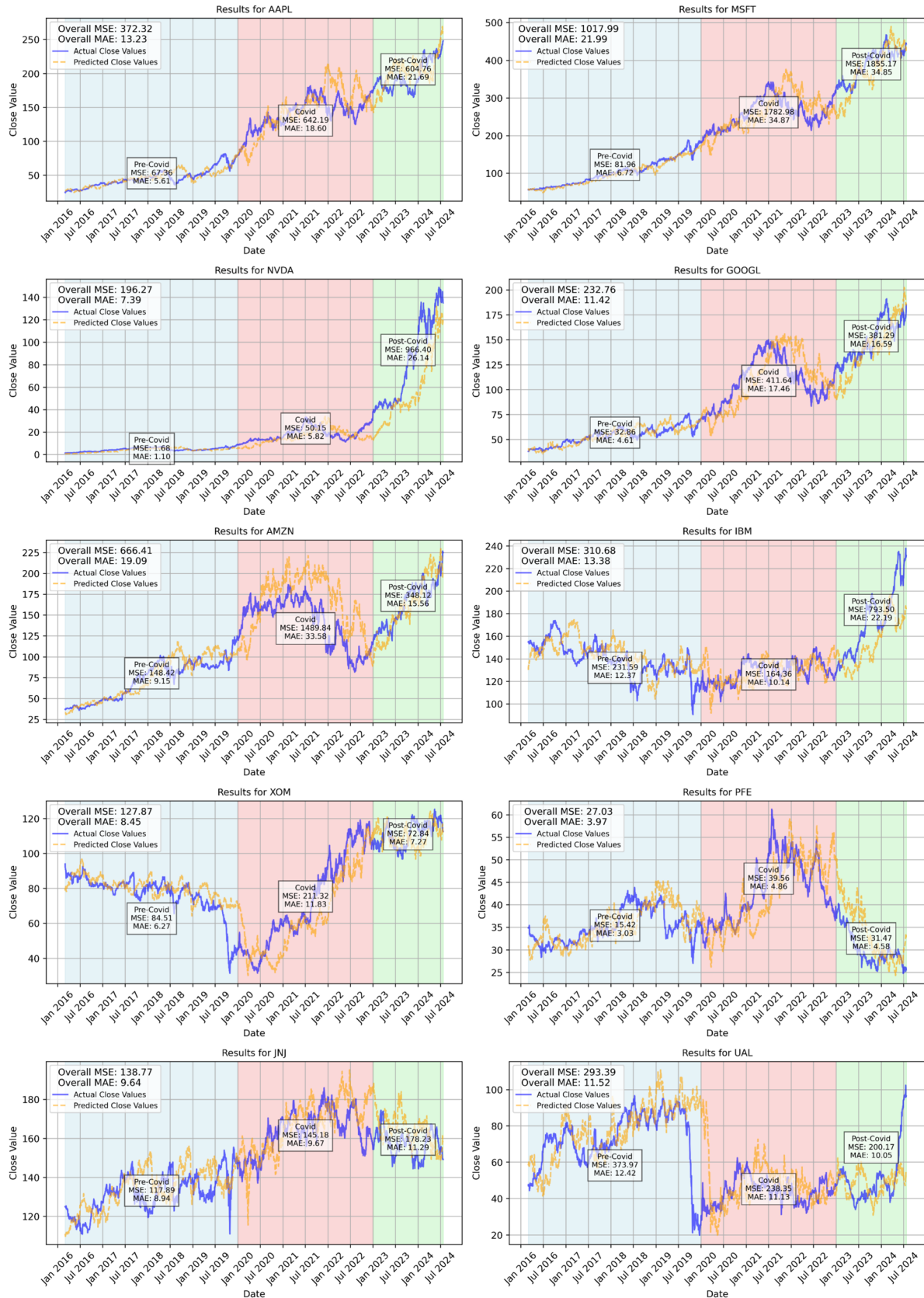
AMZN, representing the consumer discretionary sector, exhibits a different trend compared to technology stocks. The pre-COVID-to-COVID MSE increases are moderate, with less than 500% across all prediction horizons. Interestingly, the COVID-to-post-COVID transitions reveal declines in MAE, particularly for the 100-day horizon, where AMZN shows a nearly 20% reduction in prediction errors. This behavior suggests that consumer discretionary stocks like AMZN benefited from relatively higher predictability during the recovery phase, potentially driven by increased demand for e-commerce services.

XOM, representing the energy sector, shows modest increases in prediction errors compared to technology stocks. For the 1-day horizon, MSE increases are below 500%, and for longer horizons, the increases are even smaller. This sector's resilience in modeling may be attributed to its dependence on macroeconomic factors like oil prices, which exhibited more predictable recovery trends during the post-COVID phase.

Healthcare stocks such as PFE and JNJ exhibit relatively lower increases in MSE and MAE across all horizons. For example, PFE shows less than 200% MSE increases for the 1-day horizon, and JNJ reflects similar trends. These modest error changes highlight the relative stability of healthcare stocks during the pandemic, likely due to sustained demand and the sector's role in vaccine and treatment development.

UAL, representing the airlines sector, stands out due to its unique behavior. The 1-day and 10-day horizons show modest increases in prediction errors, with MSE changes remaining below 500%. However, the 100-day horizon reveals a more significant rise in MSE, reflecting the long-term uncertainty surrounding the sector's recovery. Airlines faced significant disruptions during the pandemic, including travel restrictions and demand collapse, making accurate long-term modeling particularly challenging.

Forecast Results for 100 Days Prediction Horizon



◀**Fig. 7** Forecast Results for 100-Day Prediction Horizon**Table 2** Forecasting Performance Metrics (MSE and MAE) Across Pre-COVID, COVID, and Post-COVID Periods for Different Prediction Horizons

Asset	Error	Pre-COVID			COVID			Post-COVID			Overall		
		1 Day	10 Days	100 Days	1 Day	10 Days	100 Days	1 Day	10 Days	100 Days	1 Day	10 Days	100 Days
AAPL	MSE:	0.54	5.72	67.36	8.75	69.18	642.19	8.81	90.33	604.76	5.28	46.79	372.32
	MAE:	0.52	1.82	5.61	2.22	6.76	18.60	2.21	7.71	21.69	1.50	4.86	13.23
MSFT	MSE:	2.18	11.15	81.96	26.24	189.01	1782.98	30.71	277.72	1855.17	17.04	132.77	1017.99
	MAE:	1.06	2.53	6.72	3.85	11.17	34.87	4.25	13.40	34.85	2.76	7.98	21.99
NVDA	MSE:	0.01	0.14	1.68	0.38	3.68	50.15	7.01	63.76	966.40	1.72	15.49	196.27
	MAE:	0.08	0.27	1.10	0.43	1.38	5.82	1.71	5.64	26.14	0.57	1.85	7.39
GOOGL	MSE:	0.65	4.87	32.86	4.81	36.00	411.64	7.02	68.11	381.29	3.54	29.60	232.76
	MAE:	0.57	1.69	4.61	1.61	4.77	17.46	1.91	6.40	16.59	1.24	3.82	11.42
AMZN	MSE:	1.78	14.45	148.42	11.81	112.28	1489.84	9.37	75.56	348.12	7.01	62.39	666.42
	MAE:	0.89	2.68	9.15	2.59	8.38	33.58	2.30	6.66	15.56	1.81	5.57	19.09
IBM	MSE:	3.62	34.29	231.59	5.77	51.62	164.36	5.56	67.44	793.50	4.81	47.73	310.68
	MAE:	1.34	4.45	12.37	1.67	5.37	10.14	1.57	6.24	22.19	1.51	5.17	13.38
XOM	MSE:	1.14	8.82	84.51	2.65	24.68	211.32	3.02	23.57	72.84	2.09	17.67	127.87
	MAE:	0.83	2.27	6.27	1.23	3.75	11.83	1.34	3.84	7.27	1.08	3.14	8.45
PFE	MSE:	0.19	1.70	15.42	0.66	6.92	39.56	0.26	2.89	31.47	0.37	3.80	27.03
	MAE:	0.32	0.96	3.03	0.59	2.05	4.86	0.38	1.33	4.58	0.43	1.42	3.97
JNJ	MSE:	2.02	17.83	117.89	4.36	34.15	145.18	2.65	29.50	178.23	2.98	26.16	138.77
	MAE:	0.96	3.10	8.94	1.48	4.46	9.67	1.17	4.45	11.29	1.19	3.88	9.64
UAL	MSE:	2.99	24.53	373.97	3.59	44.12	238.35	2.15	20.98	200.17	3.01	30.63	293.39
	MAE:	1.27	3.89	12.42	1.36	4.55	11.13	1.07	3.48	10.05	1.26	4.03	11.52

5.1 Statistical analysis of forecast errors across market regimes

To quantitatively assess the impact of market conditions on forecasting accuracy, a statistical analysis was conducted to compare forecast error distributions across the pre-COVID, COVID, and post-COVID periods. This analysis employs the Kolmogorov–Smirnov (KS) test to detect distributional shifts and Levene’s test to examine changes in variance, to analyze how forecast errors evolved under different market conditions.

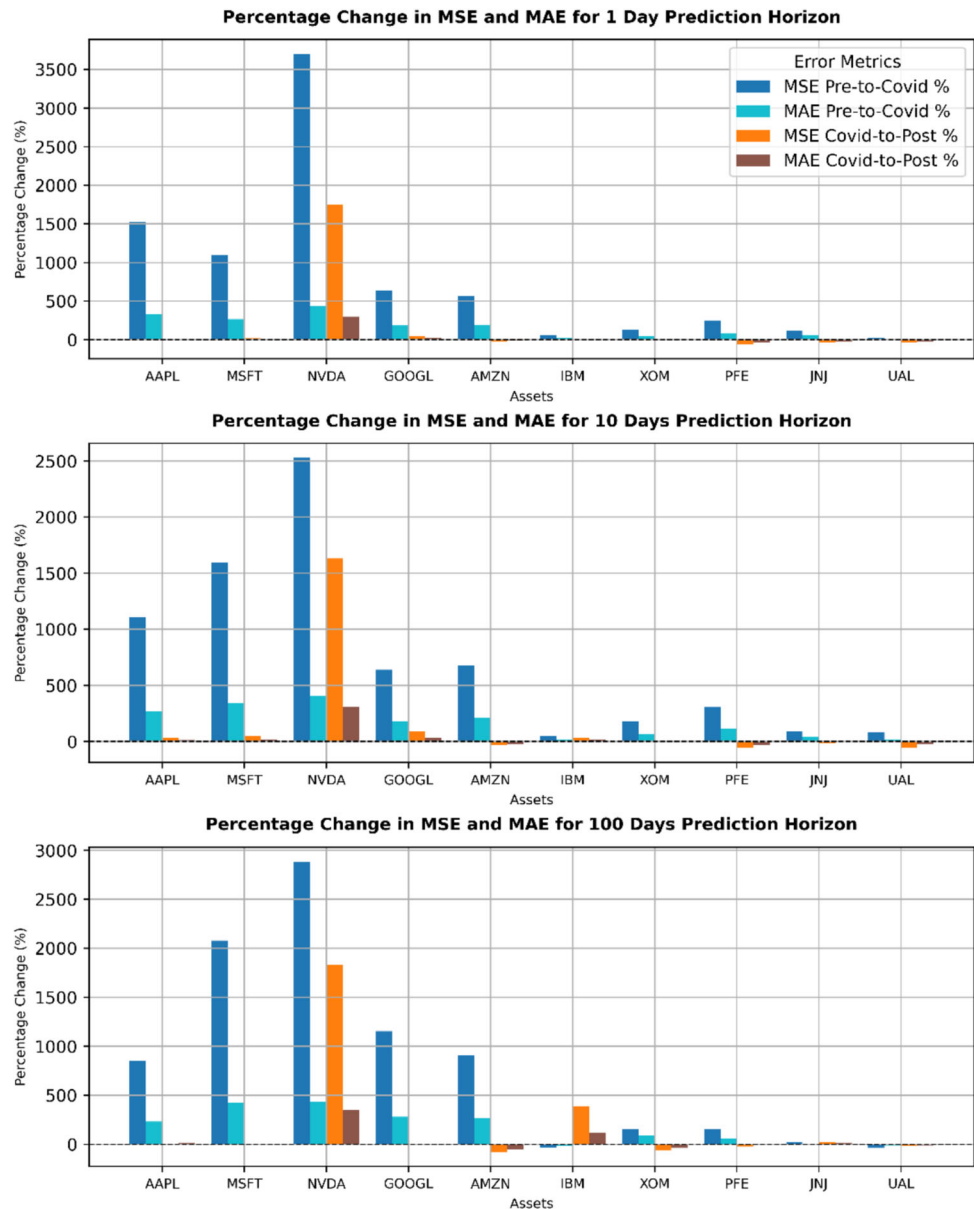
The Kolmogorov–Smirnov (KS) test is a nonparametric test used to compare two empirical distributions. Given two cumulative distribution functions, $F_1(x)$ and $F_2(x)$, the KS test statistic is defined as [72–74]:

$$D = \sup_x |F_1(x) - F_2(x)| \quad (23)$$

where D represents the maximum absolute difference between the two distributions. A small p value ($p < 0.05$) suggests that the two distributions are significantly different.

The Levene’s test is used to assess the equality of variances between two or more groups. Given a dataset split into k groups, the test statistics are computed as [75–77]:

Fig. 8 Comparison of Percentage Changes in Prediction Errors (MSE and MAE) Across COVID Periods



$$W = \frac{(N - k)}{(k - 1)} \frac{\sum_{i=1}^k N_i (\bar{Z}_i - \bar{Z})^2}{\sum_{i=1}^k \sum_{j=1}^{N_i} (Z_{ij} - \bar{Z}_i)^2} \quad (24)$$

where N is the total number of observations, N_i is the sample size of group i , $Z_{i,j}$ represents transformed absolute deviations, and $\bar{Z}_i$ and $\bar{Z}$ are the group-wise and overall means, respectively. A significant p value ($p < 0.05$) indicates that at least one group exhibits a different variance from the others. Table 3 presents the detailed statistical results supporting these conclusions. It summarizes the outcomes of Kolmogorov–Smirnov and Levene’s tests across different market regimes. Levene’s test results further support these findings by showing substantial changes in variance across market regimes.

Applying these tests to forecast errors across market regimes reveals critical insights into how prediction uncertainty evolved. The KS test results indicate significant distributional shifts from pre-COVID to COVID for all assets ($p < 0.05$), confirming that forecasting errors followed a markedly different distribution during the high-

Table 3 Statistical Analysis of Forecast Error Distributions Across Market Regimes

Asset	KS Test Pre-COVID vs COVID	KS Test COVID vs Post-COVID	KS Test Pre-COVID vs Post-COVID	Levene Test Pre-COVID vs COVID	Levene Test COVID vs Post-COVID	Levene Test Pre-COVID vs Post-COVID
AAPL	2.55E-04	9.69 E-04	8.27 E-01	4.01E-15	7.57E-13	4.06 E-01
MSFT	3.43 E-04	8.49 E-02	2.37 E-01	4.05E-18	6.62 E-08	1.93 E-02
NVDA	1.14 E-03	6.66 E-02	6.82 E-03	9.30 E-09	2.43 E-02	5.11 E-03
GOOGL	7.67 E-06	1.26 E-03	3.04 E-04	8.56E-18	1.20 E-02	9.21 E-08
AMZN	5.31 E-06	1.27 E-01	2.59 E-02	1.83 E-16	8.61 E-04	2.36 E-05
IBM	2.34 E-03	2.29 E-03	5.55 E-02	1.52 E-10	9.06 E-08	7.69 E-01
XOM	1.42 E-11	4.76 E-04	2.58 E-02	5.55E-49	2.69 E-17	6.44 E-08
PFE	2.19 E-05	6.34 E-02	0.00E + 00	6.35E-25	2.13 E-04	1.10 E-09
JNJ	5.57 E-03	7.65 E-02	1.25 E-01	1.31 E-06	2.65 E-04	6.01 E-01
UAL	1.59 E-06	2.50 E-05	5.85 E-04	5.74E-31	1.53 E-09	4.00 E-07

volatility pandemic period. The largest discrepancies were observed in technology stocks, such as GOOGL ($p = 7.66 \times 10^{-6}$) and XOM ($p = 1.42 \times 10^{-11}$), reflecting the extreme fluctuations in market conditions. In contrast, comparisons between COVID and post-COVID periods yielded mixed results. While some assets, such as MSFT and NVDA, showed no significant difference ($p > 0.05$), others (GOOGL and XOM) continued to exhibit statistically distinct error distributions, suggesting that forecasting challenges persisted beyond the pandemic.

The transition from pre-COVID to COVID resulted in a statistically significant increase in forecasting error variance for all assets ($p < 0.05$), with particularly extreme effects in NEXT_XOM ($p = 5.54 \times 10^{-49}$) and NEXT_GOOGLE ($p = 8.56 \times 10^{-18}$), indicating heightened uncertainty and error magnification during market shocks. While some assets showed a partial return to pre-pandemic variance levels in the post-COVID period, others (PFE, XOM, GOOGL) retained significantly higher variances, suggesting that forecasting uncertainty remained elevated even as markets stabilized.

5.2 Computational considerations, symbolization trade-offs, and model sensitivity analysis

The proposed methodology, which combines symbolic dynamics and graph-theoretic approaches, offers a structured and interpretable framework for analyzing nonlinear financial time series. However, its complexity and inherent limitations present challenges that require consideration.

An important consideration in the computational efficiency of the symbolization process is its $O(N)$ complexity, where N is the length of the time series. The transformation from raw numerical values to symbolic states involves three primary steps: normalization, discretization into amplitude symbols, and trend state assignment. Each of these operations is applied sequentially to the time series, requiring only a single pass through the data, ensuring that the computational cost scales linearly with the dataset size. The transition matrix construction, which follows symbolization, operates by iterating through $N - w$ overlapping windows of length w to capture state transitions, resulting in a computational cost of $O(N - w) \approx O(N)$ for large datasets where w is significantly smaller than N . Compared to traditional feature extraction methods that rely on high-dimensional technical indicators or complex handcrafted representations, symbolization remains a highly efficient alternative, preserving structural information while mitigating computational overhead.

Symbolization involves normalizing time series data and discretizing it into symbolic states. The choice of alphabet size and window length directly influences the granularity and accuracy of the symbolic representation. A comparison of these choices for the 1-day prediction horizon is presented in Table 4. A small alphabet (e.g.,

Table 4 Impact of Window Size and Amplitude Alphabet Size on Forecasting Performance (MSE) Across Different Assets

Window Size:	50 timesteps			100 timesteps			250 timesteps		
	Amplitude	Alphabet Size:		4	8	10	4	8	10
Asset	MSE	MSE	MSE	MSE	MSE	MSE	MSE	MSE	MSE
AAPL	5.39	5.27	4.53	5.23	5.14	5.28	5.45	5.31	5.22
MSFT	16.63	15.25	14.94	19.94	14.47	17.04	16.99	19.81	17.21
NVDA	1.84	1.76	1.82	1.82	8.68	1.72	2.06	2.01	2.01
GOOGL	3.79	3.68	3.59	3.77	3.55	3.54	4.15	10.56	10.21
AMZN	7.56	7.16	7.15	7.55	12.90	7.01	7.98	7.87	8.45
IBM	3.58	3.46	3.24	3.74	3.44	4.81	3.39	14.26	3.44
XOM	2.39	2.75	1.36	1.34	1.53	2.09	1.68	3.12	3.12
PFE	0.27	0.25	0.25	2.92	0.25	0.37	0.27	0.25	0.28
JNJ	2.93	2.41	24.68	2.60	2.46	2.98	11.42	2.57	11.27
UAL	3.57	7.23	3.07	3.25	2.78	3.01	3.32	3.20	3.43
TOTAL	47.95	49.22	64.63	52.16	55.2	47.85	56.71	68.96	64.64

3–5 symbols) leads to information loss, whereas a large alphabet (e.g., more than 10 symbols) increases noise sensitivity and computational cost. The results indicate that an alphabet size of 10 consistently outperforms smaller alternatives across multiple assets, including AAPL (4.53 MSE), MSFT (14.94 MSE), and AMZN (7.15 MSE) when using a 100-timestep window. These findings suggest that an alphabet size of 10 balances granularity and robustness, capturing essential signal variations while avoiding excessive noise amplification.

Similarly, window size plays a crucial role in capturing the temporal dependencies of financial time series. A key aspect of the amplitude symbolization process is normalizing time series data before symbol assignments. Each windowed segment of length w undergoes min–max normalization, ensuring that the transformed values lie within a fixed range $[0,1]$. This transformation effectively reduces the impact of extreme outliers, preventing high-magnitude fluctuations from dominating the symbolic representation. However, the effectiveness of normalization is inherently dependent on window size. A shorter window (e.g., 50 timesteps) may fail to capture broader trends and remain susceptible to local spikes. In contrast, an excessively long window (e.g., 250 timesteps) may lead to over-smoothing, suppressing meaningful variations, and distorting high-frequency components essential for short-term predictions.

Furthermore, the total MSE values across all assets confirm the superiority of this configuration. The 100-timestep window with 10 symbols yields the lowest total MSE (47.85), significantly outperforming the 50-timestep (64.63) and 250-timestep (64.64) alternatives. This highlights that a 100-timestep window effectively balances short-term and long-term dependencies, enabling the model to achieve accuracy and computational efficiency. The increased errors in the 50-timestep window indicate that shorter contexts fail to encode sufficient information. At the same time, the degradation in performance at 250 timesteps suggests that an overly extended context reduces the model's sensitivity to critical short-term fluctuations.

The construction of state transition matrices offers a potent probabilistic framework for capturing temporal dynamics, providing a valuable understanding of the underlying structure of financial time series. While it primarily assumes first-order Markov properties, this approach effectively models short-term dependencies and can be extended to incorporate higher-order transitions to capture more complex patterns. However, the efficacy of these transition matrices in deep learning models heavily relies on robust preprocessing layers that ensure consistency, noise reduction, and enhanced feature extraction before being fed into neural architectures.

To assess the impact of different architectural choices, an ablation study is conducted by systematically removing key components and measuring their effect on forecasting performance. The following inferences are obtained:

- The inclusion of Conv2D layers significantly enhanced short-term predictions by capturing localized symbolic dependencies in state transition matrices. The removal of these layers led to a notable increase in error variance, indicating that spatial patterns in symbolic sequences are crucial for precise forecasting.
- While LSTMs demonstrated superior stability and predictive accuracy in most cases, GRUs performed comparably for certain assets, XOM and PFE, as indicated by their lower MSE values in Table 5.
- The proposed combination, incorporating both Conv2D and LSTM layers, yielded the most consistent accuracy across assets, reinforcing the importance of capturing both local patterns and long-range dependencies in financial time series through transition matrices.

The comparison results, summarized in Table 5, highlight the influence of each architectural component on forecasting performance. These findings emphasize the necessity of Conv2D layers for capturing local features, LSTM layers for maintaining long-range dependencies.

While the proposed approach provides a computationally efficient alternative to direct deep learning-based methods, certain trade-offs must be acknowledged. The computational complexity of symbolization and transition matrix construction is $O(N)$, making it scalable for large datasets, yet the additional storage and processing of symbolic states add complexity compared to traditional continuous-valued time series models. The hierarchical stacking of transition matrices ensures that both short- and long-range temporal dependencies are captured, but this also introduces potential sensitivity to the choice of symbolic parameters, including alphabet size and window length. Moreover, while the proposed approach simplifies feature extraction and enhances interpretability, it requires careful tuning to optimize performance across different market conditions.

5.3 Rolling expanding window cross-validation

While the 50–50 temporal split serves as the primary validation strategy in this study, rolling expanding window cross-validation is applied on 1-day prediction horizon model as a complementary method to assess how forecasting performance varies across different temporal segments. This approach not only supports the validity of the 50–50 split but also offers insights into the stability and adaptability of the models when exposed to gradually increasing training periods [78, 79]. This method progressively expands the training window while shifting the test set forward in time. The rolling window cross-validation approach is implemented using a fixed window size for training and testing:

- The training set starts with 10% of the data.
- Test set is the next 10% of the data immediately following the training set.
- The training set expands by incorporating the previous test set, while the test set moves forward by another 10%.
- This process is repeated until the entire dataset is covered, leading to multiple validation splits.

Table 5 Systematic Ablation Studies on the Individual Contributions of Architectural Components

Condition: Asset	Without Conv2D Layers MSE	Without LSTM Layer MSE	LSTM Replaced with GRU Layer MSE	Proposed Combination MSE
AAPL	88.72	5.14	119.52	5.28
MSFT	20.12	16.12	15.00	17.04
NVDA	8.08	1.78	5.59	1.72
GOOGL	8.73	3.60	10.57	3.54
AMZN	11.90	7.40	12.32	7.01
IBM	8.76	3.68	13.32	4.81
XOM	1.32	1.40	1.47	2.09
PFE	0.67	0.26	0.35	0.37
JNJ	18.18	15.17	12.37	2.98
UAL	5.06	2.88	3.06	3.01

This ensures that each model is trained only on past data while forecasting the subsequent period, making it a realistic evaluation method for financial time series forecasting. This progressive training expansion allows us to evaluate:

The effectiveness of different training period lengths in capturing essential patterns for forecasting.

The impact of temporal variations on model performance, particularly in response to structural changes in the data.

Potential performance trends that may emerge when the model is exposed to longer training histories.

By implementing this cross-validation scheme, fluctuations in forecasting accuracy across different temporal splits are analyzed. This examination is crucial in financial markets, where abrupt shifts in market conditions can influence model reliability. Table 6 presents the mean squared error (MSE) results for the rolling expanding window cross-validation across different train–test splits for multiple assets. Several key observations emerge:

- Early-stage models (10–50% training windows) exhibit relatively stable forecasting performance, indicating that short-term historical data are sufficient for certain assets but may be insufficient for others.
- Performance variations in later windows (60–100%) suggest that some assets experience greater volatility or non-stationary effects that impact forecasting accuracy.
- Longer training periods (e.g., 50–100%) do not universally guarantee better results, indicating that optimal training window selection remains asset-dependent.

These findings reinforce the rationale for the 50–50 split as a robust validation approach while also highlighting the nuances of forecasting performance across different temporal horizons.

6 Conclusion

This study advances the theoretical and empirical understanding of financial time series forecasting by integrating symbolic dynamics with a convolutional recurrent neural network (CRNN) architecture. Symbolic dynamics provides a powerful preprocessing framework that transforms continuous-valued stock price data into sequences of discrete symbols, preserving essential structural patterns and emphasizing qualitative behaviors. Discretizing the data into amplitude and trend states, followed by the construction of state transition matrices, enables the representation of the probabilistic structure of market movements. The methodology uniquely focuses on time series data derived exclusively from economic factors, such as stock prices, trading volumes, and volatility, without incorporating external political or social variables. This self-contained framework highlights the strength of symbolic dynamics in capturing inherent market fluctuations while filtering out noise. By discretizing time series data into symbolic states based on amplitude and trend, and representing their transitions probabilistically,

Table 6 Rolling Expanding Window Cross-Validation Results

Train Period (%)	Test Period (%)	AAPL MSE	MSFT MSE	NVDA MSE	GOOGL MSE	AMZN MSE	IBM MSE	XOM MSE	PFE MSE	JNJ MSE	UAL MSE
0–10	10–20	0.01	0.11	0.00	0.05	0.02	0.74	0.28	0.02	110.43	0.42
0–20	20–30	0.06	0.12	0.00	0.07	0.07	1.97	0.49	0.02	0.20	0.49
0–30	30–40	7.65	0.21	0.00	0.13	0.10	1.89	0.30	0.04	0.31	0.84
0–40	40–50	0.20	0.51	0.00	0.29	0.22	1.83	0.60	0.07	357.56	2.62
0–50	50–60	0.19	0.45	0.01	0.22	0.29	1.43	0.33	0.06	899.50	1.94
0–60	60–70	0.77	6.28	0.22	12.19	3.25	2.38	0.60	0.16	15.19	3.93
0–70	70–80	6.57	20.63	0.13	3.23	10.01	5.28	1.23	0.32	4.55	5.80
0–80	80–90	10.74	31.85	0.72	7.75	14.92	3.09	3.01	0.71	3.15	2.10
0–90	90–100	9.00	31.36	9.61	7.78	11.65	7.40	2.33	0.20	2.15	2.32
0–50	50–100	5.28	17.04	1.72	3.54	7.01	4.81	2.09	0.37	2.98	3.01

the methodology emphasizes structural and qualitative patterns that are crucial for analyzing nonlinear and volatile financial domains.

From a theoretical standpoint, the combination of symbolic dynamics with CRNN-based forecasting aligns with principles from ergodic theory, information theory, and Markovian dynamics. Representing complex temporal dependencies in a compact, state-based manner reduces the problem's dimensionality while preserving its nonlinear characteristics. The symbolic-state transitions are then encapsulated in tensors suitable for deep learning models, where convolutional layers focus on extracting spatial patterns within transition matrices and recurrent layers capture temporal dependencies spanning multiple time windows.

Empirically, the results underscore the model's adaptability across varying market regimes and forecasting horizons. During the stable pre-COVID period, the CRNN achieved relatively low error metrics, with MSE and MAE values as low as 0.01 and 0.08 for NVDA's 1-day horizon. The reduced volatility in this phase allowed the symbolization and neural architecture to exploit well-defined market patterns. However, as market uncertainty intensified during the COVID period, both short- and long-term forecasts became more challenging, reflected in increased errors, e.g., MSFT's 1-day horizon MSE surged to 26.24, and AMZN's 100-day horizon MSE reached 1489.84. These results highlight the intrinsic difficulty of forecasting in crisis conditions and reinforce the value of symbolization as a robust baseline technique, even when facing abrupt regime changes.

Post-COVID, moderate improvements in predictive accuracy emerged as markets began stabilizing, though they remained more volatile than pre-COVID conditions. For instance, AAPL's 100-day horizon MSE decreased from 642.19 during COVID to 604.76 post-COVID. These changes emphasize the necessity of adaptive models that not only learn from historical stable regimes but also accommodate and recalibrate as new structural patterns emerge.

The multi-horizon forecasting experiments demonstrate a trade-off between forecast length and accuracy. While 1-day predictions often achieved high precision, longer-term horizons amplified cumulative uncertainties. Nevertheless, the use of symbolic dynamics, normalized transition matrices, and CRNN architectures enabled the retention of meaningful information, yielding competitive performance even for extended horizons like 100 days.

While the 50–50 temporal split was adopted as the primary validation strategy to ensure a stable and unbiased assessment of forecasting accuracy, the rolling expanding window cross-validation provided further acuties into the variations in model performance across different time horizons. The observed fluctuations in error metrics, MSE, across expanding windows emphasize the importance of selecting an appropriate training horizon that balances historical representativeness and adaptability to structural changes. The results confirm that while a fixed validation split offers a reliable benchmark, additional cross-validation approaches can help reveal forecasting sensitivities to different market conditions.

Despite its promising results, the methodology's exclusive reliance on time series data from economic factors represents both a strength and a limitation. On the one hand, this focus ensures scalability and independence from external, less predictable factors like political or social events. On the other hand, it overlooks potentially valuable contextual information from other domains that could enhance forecasting performance. For instance, incorporating macroeconomic indicators, geopolitical developments, or consumer sentiment data through data fusion approaches could provide a more comprehensive understanding of market dynamics and improve the model's responsiveness to external shocks.

The growing adoption of graph-theoretic approaches in financial forecasting has further highlighted the need to model complex interdependencies among financial assets. Traditional time series models often overlook structural relationships between market entities, whereas graph neural networks (GNNs) provide a robust framework for capturing dynamic interactions, sectoral dependencies, and global market influences [40–42, 45–50]. Recent advancements in GNN-based forecasting have demonstrated superior performance in modeling stock co-movements, leveraging adjacency matrix representations and message-passing architectures to uncover latent structural patterns in financial networks [46, 48, 50]. By integrating graph-based methodologies with symbolic transition matrices, future research can further enhance interpretability, exploit higher-order relationships, and develop more adaptive forecasting models that effectively respond to evolving market dynamics.

As a final note, while the proposed framework demonstrates robustness and adaptability across diverse financial scenarios, future research can build upon these findings by integrating symbolic dynamics with data fusion techniques. By incorporating complementary information from political, social, and economic domains, as well as refining symbolization schemes and enhancing neural architectures with attention mechanisms or explainable AI frameworks, the methodology can be further strengthened. Such advancements will not only improve forecasting accuracy in volatile and complex environments but also provide a more holistic approach to understanding financial market behaviors, ultimately guiding better investment decisions and risk management strategies.

Author contributions Fuat Kaan Mirza designed the study, performed the analysis, and wrote the manuscript. Önder Pekcan contributed to construction of methodology and the interpretation of the results. Mustafa Hekimoğlu provided critical feedback and edited the manuscript. Tunçer Baykaş oversaw the research process and approved the final manuscript. All authors reviewed and approved the final version.

Funding Open access funding provided by the Scientific and Technological Research Council of Türkiye (TÜBİTAK). This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability The data used in this study were sourced from Yahoo Finance, a publicly accessible platform for historical and real-time financial data. All datasets utilized in the analysis, including stock price data and related market indicators, are openly available through the Yahoo Finance website (<https://finance.yahoo.com>). Researchers can access the same datasets by retrieving historical data for the relevant assets and time periods through the platform's data export feature.

Declarations

Conflict of interest The authors declare that they have no competing interests.

Ethical approval The submitted work is original and has not been published elsewhere in any form or language. The results/data/figures in this manuscript have not been published elsewhere, nor are they under consideration (from you or one of your contributing authors) by another publisher.

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